

SPOT-OD: Adaptive-Filter and Observability Mechanism Findings from a Simulator-Bound Orbit-Determination Audit

Authors omitted for review

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Abstract

This paper studies estimator screening for sparse-visibility orbit determination in a compact two-body+ J_2 +drag simulator. Its primary result is a mechanism diagnosis: under force-model mismatch, adaptive UKF inflates effective measurement noise and damps the state corrections needed to follow persistent dynamics bias. On the primary compact slice, adaptive UKF reaches mean effective measurement-scale inflation 3.32, while the corresponding innovation-diagnostic medians are 4.73 for adaptive UKF, 1.60 for EKF, and 1.83 for UKF. The drag-scale cascade then separates three distinct binding constraints: EKF linearisation error, sparse-geometry observability limits, and candidate ballistic-coefficient inertia.

The learned-estimator result is secondary and exploratory: across the reported internal frozen-rule replications, no evaluated learned construction beats the best tuned classical reference on observed-step position RMSE. Public LAGEOS CRD/SP3 readouts are retained only as support for the measurement pipeline and transfer boundary; one predeclared prospective public week records a numerically lower learned residual UKF mean under the full correction stack, with confidence interval spanning zero, so that readout is indeterminate rather than simulator validation or operational POD.

The submission package is organized as an inspectable worked case-study record. It ties each manuscript-facing claim to fixed rules, paired resampling, practical-significance floors, archived evidence records, a deterministic submission-validation pass, archive-extracted table regeneration and public-slice rerun checks, a minimum-tier manifest/digest audit, and one bounded learned-training replay. These checks strengthen inspection and rerun readiness, but they are not claimed as full scientific reproduction. The paper therefore contributes a reviewer-auditable compact-screening record for deciding which OD ideas merit escalation to higher-fidelity or real-data validation.

Keywords: orbit determination; satellite state estimation; sparse visibility; graph neural networks; learned filtering; claim-audited evaluation; adaptive filtering; negative results; reproducibility

1 Introduction

Contribution in one sentence. The central result is a compact-model AUKF mechanism diagnosis: under force-model mismatch AUKF inflates effective R and damps needed corrections, while the drag-scale cascade isolates EKF linearisation, sparse-geometry observability, and candidate-inertia failure modes; the learned-estimator negative is a secondary exploratory screening result, and the audit record is presented as a worked case study rather than a validated transferable framework.

Satellite OD is grounded in orbital dynamics, measurement modelling, and recursive statistical estimation [25, 79, 78, 50, 77], with Kalman-filter families and their nonlinear extensions central because they encode physical priors, uncertainty propagation, and measurement-update structure [30, 28, 29, 82]. Sparse visibility — intermittent contact, unfavourable geometry, and shifting noise statistics — motivates learned residual or adaptive components, but it also raises the evidentiary bar for any claim that a neural estimator improves on a strong dynamics-aware baseline, which is the claim audit this study enforces.

Scope and claim boundary. Conclusions are drawn strictly from the reported tables and figures. Table 2 is the governing claim–evidence–limit hierarchy, with the compact AUKF mechanism result as the supported primary claim. Observed-step position RMSE is the primary endpoint; all-step trajectory RMSE is retained only as a propagation-dominated reference because roughly four-fifths of evaluated steps have zero visible

Table 1: Main-text abbreviation guide for the estimator and audit terms used in the Results.

Term	Meaning	Term	Meaning
OD/POD	orbit determination / precise orbit determination	EKF/UKF/AUKF	extended, unscented, and adaptive UKF filters
PUKF	predeclared process-noise adaptive UKF	DMC-EKF	dynamic-model compensation EKF with empirical acceleration
DSA-EKF/UKF	drag-scale adaptive EKF/UKF	RGR-GF	residual graph recurrent estimator with graph filtering priors
DBAR	dynamics-bias adaptation-risk heuristic	NIS/CRLB	normalized innovation squared / Cramer-Rao lower bound
SGP4/TLE	public-catalog propagator / two-line element set	SLR/SP3	satellite laser ranging / precise-orbit state product
CRD/ILRS	Consolidated Laser Ranging Data / International Laser Ranging Service	CDDIS	NASA/CDDIS SLR data archive
WLS	weighted least-squares offline batch OD reference	RFIS/VG-RFIS	offline smoother and visibility-gated composite; see supplement glossary
IDP-RGR-GF	identity-prior residual variant; see supplement glossary	Scope-only	retained for scope, stress, or provenance interpretation; not used for pass/fail conclusions or headline claims
V&V	verification and validation; used here as adjacent credibility lineage		

stations. The evidence is simulator-bound, internally fixed before the claimed runs rather than externally registered, and bounded to the tested constructions and compact sparse-visibility regimes. The learned estimators covered here are the authors’ bounded residual constructions plus one validation-tuned KalmanNet-style learned-gain comparator; the public CRD/SP3 material is support-only measurement-pipeline evidence, not a co-equal empirical headline. The compact-physics ceiling also matters: a negative weighs against advancing the tested claim beyond this compact prior, sparse geometry, and candidate set, but it is not a claim that learned OD corrections cannot help under a different force model, sensor suite, or public-reference regime. For the full exclusions and limitations, see Table 2 and Section 7.

Contributions. *(C1) Adaptive-filter mechanism findings:* OD-specific quantification and diagnostic decomposition of the known covariance-matching failure mode — under force-model mismatch in a compact two-body+ J_2 +drag OD simulator, AUKF inflates effective R (mean 3.32; median R -only NIS 4.73 versus 1.60/1.83 for EKF/UKF) and damps state corrections; the drag-scale cascade separates EKF linearisation failure, sparse-geometry observability limits, and candidate-side β -prior inertia as distinct binding constraints, showing median trajectory-mean $\hat{\beta} = 1.0000$ despite 9.6 km median arc-mean drag separation at enriched geometry. A secondary mechanism finding is the long-arc EKF/AUKF ordering reversal: on 3-hour higher-fidelity arcs, AUKF has lower RMSE than EKF on the tested long-arc slice (EKF–AUKF mean +264.4 m, CI strictly above zero), showing that compact-model AUKF prescription is regime-bound. These mechanism decompositions inform estimator screening for OD campaigns: before committing to precise-reference validation, learned candidates are screened against compact-simulator gates testing tuned adaptive guardrails (AUKF), offline batch-OD references, frozen endpoints under the observed-step hierarchy, practical-significance floors, and structural-channel alternatives (DMC-EKF, DSA-EKF).

(C2) Worked inspection record: the evaluation discipline (trajectory-unit paired resampling within declared OD slices, separate noise-side and force-side structural-channel withdrawal rules, and a shared observed-step visibility mask across comparators) is documented as a worked, inspectable case-study record under self-authored definitions, not as a validated framework transferable beyond this study.

(C3) Exploratory bounded learned-negative (secondary): under the primary observed-step endpoint, no evaluated learned construction beats the per-scenario best tuned classical reference in the $K = 32$ frozen-rule replication, and the PUKF/DMC-EKF/DSA-EKF variants and DBAR heuristic likewise miss their predeclared criteria. The endpoint was chosen after the training-cohort review, and the $K = 32/K = 96$ checks are internal frozen-rule replications rather than external preregistration, so this remains a bounded screening result that still needs prospective external confirmation.

Aerospace relevance. Sparse OD evidence is a campaign-design problem as much as an estimator-ranking problem. Acta-style OD claims depend on geometry, dynamics fidelity, and operational context. Recent studies span operational and public-data OD contexts [7, 55, 44, 46, 43], alongside learned and adaptive

filter applications [91]; SPOT-OD addresses learned-estimator evaluation rather than operational public-orbit maneuver reconstruction. A practitioner takeaway: before committing to a precise-reference reduction or operations-facing validation campaign, an estimator-selection gate can require learned candidates to survive tuned adaptive guardrails, frozen endpoints, practical-significance floors, and provenance checks on a compact simulator. Learned estimators that fail these checks under compact dynamics remain pre-campaign hypotheses rather than validated upgrades; those that pass become candidates for higher-fidelity follow-up. SPOT-OD provides the compact-simulator gate mechanics: apparent positives are supported, bounded, demoted, or withdrawn by predeclared rules rather than post-hoc narrative.

Estimator family evaluated under the self-audit record. The record covers tuned EKF/UKF/AUKF baselines, an offline robust batch WLS OD reference, the authors’ bounded residual learners, a validation-tuned in-house KalmanNet-style learned-gain comparator (S-Table 78), and three predeclared dynamics-bias responses: PUKF, DMC-EKF, and DSA-EKF. The in-house learned-gain comparator anchors the learned-negative alongside the RGR-family constructions. The bounded learned-negative is not solely based on author-defined RGR residual variants: it is accompanied by external-family boundary checks including a canonical untuned Ensemble Kalman Filter (EnKF) comparator. The main text retains only summary learned-negative evidence and the headline $K = 32$ replication; detailed learned-comparator diagnostics (including the $K = 96$ stress replication, EnKF comparator, and structural-channel checks) are housed in the supplement to preserve the central C1 mechanism finding’s visibility.

What survives scrutiny. Reviewers can verify five bounded claims: (1) the compact AUKF R -inflation mechanism under force-model mismatch (Table 5, Figure 1); (2) the drag-scale cascade that localizes the binding failure modes; (3) the secondary learned-negative across $K = 32/K = 96$ internal frozen-rule replications (Table 4, S-Tables 17, 19); (4) DBAR withdrawn plus support-only public CRD/SP3 measurement-pipeline readouts reported with explicit scope limits (Section 5.13; S-Table 56); and (5) an inspectable evidence record tying seeds, rules, evaluations, and validation artifacts to claims (see Table 2 for scope boundaries). Confidential package: fixed settings, rule/estimator/evaluation records, digests, and validation reports. Deeper reruns (documented, not completed): full training, filter recomputation, live data retrieval.

Audit outputs (bounded negatives). Every output is bounded to the authors’ constructions (Table 2). No learned construction beats classical on $K = 32$ observed-step; $K = 96$ stress preserves the negative at floor-power scale. DMC-EKF is not better than UKF (CI $[-3.8, +8.1]$ m); DSA-EKF is below its practical floor. The 3-hour long-arc replication confirms the DSA-EKF negative: DSA-EKF is worse than AUKF (DSA-EKF–AUKF CI $[+8.7, +697.9]$ m). DBAR is withdrawn after its 450-realization characterization is statistically indistinguishable from an 81.3% no-information baseline. See supplement S-tables for full details.

Measurement-geometry observability diagnostic cascade on the drag-scale structural channel. A three-slice *measurement-geometry observability diagnostic cascade* on the augmented multiplicative drag-scale parameter β is reported in the Results after the structural-channel estimators and decision predicate have been introduced, with full details in the supplement (supplementary Sections 6–8). Each slice fixes the drag-scale channel as the structural form matched to the synthetic bias, freezes the candidate hyperparameters from the prior validation selection, and applies the predeclared structural-channel decision predicate unchanged; all three slices return predeclared bounded structural-channel negatives. The cascade is reported as a mechanism diagnosis, not as a calibration positive for the self-audit record.

2 Related Work and Positioning

Orbit propagation and statistical estimation. The study is grounded in the SGP4/NORAD lineage and standard astrodynamics references [25, 79, 78, 50], with statistical OD providing the framework for combining dynamics, measurements, and uncertainty [77]. This motivates a baseline-first stance: learned methods must be evaluated against strong physical filters, not weak comparators.

Recursive filters and learned filtering. The Kalman filter [30], EKF [28] and the unscented transform [29, 82] are standard recursive estimators. Adaptive Kalman methods motivate innovation/smoothing-based noise adjustment; learned filtering, including KalmanNet, motivates hybrid designs [48, 34, 21, 64, 52]. An independent physics-informed ML cislunar OD study reinforces that learned-OD evaluation must weigh calibration and uncertainty rather than point accuracy alone [1]; deep-neural nonlinearity detectors driving order-switching filters are complementary neural-aided designs [92]. Improvements over fixed-noise UKF can reflect noise-adaptation already represented by AUKF. Classical adaptive filtering recovers noise statistics only when the model structure is correct [48, 65, 28]; robust Kalman work on heavy-tailed noise is adjacent context, not a direct OD baseline [84]. The AUKF result reported here instantiates the known covariance-matching limitation under model mismatch established in the classical adaptive-filter literature: when the dominant residual is dynamics or force-model error rather than noise-distribution error, adaptive noise-statistics adjustment can degrade rather than improve estimation [48, 65, 28]. The OD-specific quantification here is that AUKF inflates effective R under compact-model force-bias in sparse-visibility OD. Chihabi & Ulrich (2025) provide recent angle-only adaptive OD context, confirming that learned and adaptive filter performance depends on the source of residual mismatch (dynamics-side bias versus noise-side distributional error) [11]. When the dominant residual is dynamics/force-model error, statistical OD prescribes *dynamic model compensation* (DMC): state augmentation with low-frequency empirical-acceleration estimates to absorb unmodelled accelerations through process-model channels rather than noise reweighting [77, 86, 75, 83]. The force-mismatch and structural-channel results reported below are controlled diagnostics consistent with this lineage.

Recent learned and adaptive OD studies (2023–2026) cover: systematic LEO precise-OD reviews [70]; physics-informed neural OD, optical IOD/tracklet-correlation, and learned/adaptive filters across ground-based, SSA, cislunar, deep-space, relative-navigation, and continuous-maneuver regimes [68, 67, 62, 93, 90, 61, 18, 76, 87, 37, 91, 9, 94]; operational orbit-recovery, drag-estimation, TLE refinement, hybrid SGP4 error-learning, and continuous-maneuver OD [10, 46, 3, 69, 45]; cislunar IOD, optical-flow IOD, ISL observability, multi-scale initializers, and TLE global-optimization [73, 23, 54, 74, 55, 38, 95, 44]; and cislunar PGM/hybrid filtering, orbit-attitude/sensor-tasking coupling, and real-spaceborne GNSS adaptive EKF [56, 57, 58, 13, 35]. Recent 2026 learned-orbit predictors [39, 88, 47] address propagation/orbital-parameter forecasting rather than recursive state-estimation self-audits under fixed force models and sparse visibility. These complementary studies report estimator performance in their own regimes; SPOT-OD asks what evidence composition screens learned or adaptive improvement claims. SPOT-OD does not propose a new estimator; it places learned residual methods, noise-adaptive AUKF/PUKF, structural DMC-EKF, public replay, and selector tasks under one auditable evidence boundary. Domain-adjacent neural orbit/state-estimation and spacecraft-GNN work [80, 33] shows learned satellite inference is not new, without making those papers direct baselines here.

Worked multi-ingredient self-audit record. The seven ingredients (predeclaration, paired bootstrap, Holm/BH multiplicity control, capacity-matched controls, upstream architecture sanity reproduction, astrodynamics-grounded floors, evidence-record traceability) are individually standard practice in adjacent literatures [51, 5, 14, 60, 71, 49, 72, 89]. This study instantiates them jointly as one worked OD-specific claim-audit evidence package under self-authored definitions. The supplement provides a scoped visible-record scoring audit under self-authored definitions (S-Table 10, supplementary Section 11), not an absence proof and not independent validation. S-Table 15 is a subset-sufficiency audit showing, ingredient by ingredient, that omitting a named ingredient would have admitted a misleading positive or unsupported upgrade in this evidence set. The result is a worked auditable record for this compact sparse-visibility study, not a validated reusable framework or demonstrated cross-domain transfer. Reuse in GNSS-LEO, optical, SLR, or operations-facing OD would require rebuilding the ingredients around that domain’s measurements, force model, reference products, baselines, and predeclared gates, followed by independent application by a party other than the record designer.

Positioning vs operational POD/SLR practice and recent learned satellite estimators. Operational precise OD on satellite-laser-ranging targets routinely reaches few-centimetre orbit accuracy at international analysis centres, with a full force model, a strict ITRF $\leftrightarrow$ GCRF reduction ingesting the IERS Earth-orientation series, precise tropospheric and centre-of-mass corrections, and dedicated ranging campaigns coordinated through the ILRS [59, 27, 53, 26]. Recent operational SLR/POD studies further demonstrate

Table 2: Condensed claim–evidence–limit hierarchy for SPOT-OD. The rows separate supported claims, decision effects, diagnostic evidence, public measurement-pipeline probes, exclusions, and reproducibility support so that support material is not mistaken for operational validation.

Evidence role	Paper-facing status	Boundary
Supported primary claim	The larger frozen-rule observed-step replication is the central simulator anchor; endpoint-support and repeated-seed records support the same ordering, and the stress-only K=96 replication exceeds the floor-power design check. No evaluated learned construction beats the per-scenario best classical reference.	Compact simulator conclusion only; not external preregistration, public precise-reference evidence, operational validation, or broad learned-OD refutation.
Self-audit decision effect	The protocol-subset sufficiency audit records one misleading positive or unsupported upgrade blocked by each ingredient in this evidence set.	Retrospective sufficiency evidence for this worked OD self-audit record, not mathematical necessity, third-party validation, or universal optimality.
Bounded negative/diagnostic evidence	PUKF, DMC-EKF, and DSA-EKF fail their predeclared sparse-visibility positive criteria; a canonical untuned ensemble Kalman filter does not displace the classical baselines on any slice; DBAR is withdrawn as a claimed positive and retained/reported as a characterized bounded negative; KalmanNet-related material is retained as bounded feasibility and sanity evidence.	Bounded to these tested constructions and slices; not a general impossibility result for structural estimation or learned OD.
Public measurement-pipeline probes	Public ILRS SLR measurements and independent SP3-c products exercise parsing, reduction, state scoring, traceability, breadth, and held-out DBAR/calibrator negatives. Scope-only/support-only constructions include SGP4 multi-revolution benchmark, SatNOGS time-aligned replay, public SP3/ILRS breadth/provenance probes, and KalmanNet transposition feasibility diagnostics.	Measurement-pipeline and provenance evidence only; reduces data-handling risk for future validation but does not validate the simulator conclusion, centimetre SLR, or operational POD.
Not claimed	The manuscript does not claim a new estimator, operational POD or flight readiness, precise centimetre-level SLR validation, broad learned-OD refutation, empirical transfer to future OD pipelines, or coverage of tuned, inflated, or localized EnKF or particle/Gaussian-mixture OD methods beyond the single canonical untuned-EnKF boundary check.	These are exclusions that bound the claim hierarchy, not deferred positives.
Reproducibility support	Inspection materials separate digest and record consistency checks, generated-table checks from materialized records, one bounded learned-estimator training replay, and a full-rerun route.	Supports confidential journal-submission inspection; no DOI or public identifier is claimed at initial submission, and full raw-data/training/filter/live-data rerun is not claimed.

that precision depends on fusion strategy, range-bias handling, and prior constraints [63, 36]. The public precise-reference probes here share inputs and reference products with that practice but evaluate *compact*-dynamics estimators (two-body+ J_2 , optional drag/third-body/SRP) under sparse-visibility recursive filtering; the resulting tens to hundreds of metres magnitudes are *not* an operational precise-OD result and are not comparable to centimetre-class ILRS analyses. Their constructive role is narrower: they reduce measurement parsing, frame-reduction, state-scoring, and provenance risk before any later public-reference validation study. SPOT-OD does not propose a new estimator but reports a disciplined, predeclared negative on the authors’ own residual constructions, leaving the comparison against any faithfully reproduced published learned-OD system explicitly open.

Graph neural networks, calibration, and observability. GNNs encode relational inductive biases through message passing [32, 22, 81, 2, 19, 66]; the SPOT-OD ablation evaluates message passing variants under capacity and input matched controls but does not isolate them as the unique source of improvement (Section 6; *graph message passing is not a claimed contribution* canonically there). Adjacent SSA work has also framed large-scale resident-space-object conjunction screening as a GNN link-prediction/screening problem, which is useful context but not a direct OD estimator baseline here [12]. Calibration analysis [20] is used only as a diagnostic for the evaluated runs, not a deployment guarantee. Sparse OD is also an information problem; classical OD treats geometry and conditioning as central [77], and prior space object observability work motivates explicit geometry analysis [17]. The finite difference geometry summaries are not a formal observability proof.

3 Data Sources, Evaluation Scope, and Protocol

3.1 Archived data sources and scenario scope

The public-data layer uses archived CelesTrak and SatNOGS snapshots rather than live API state. CelesTrak general-perturbation/TLE data provide public orbital element evidence [31]; SatNOGS network, observation, and API metadata provide public ground-station and observation context [41, 42, 40]. The retained records identify source URLs, fetch timing, record counts, and digests so replay scenarios remain fixed. Primary accuracy tables use the nominal and stress synthetic splits. Public-catalog SGP4 replay is a *bounded public-catalog replay/reference slice* rather than a discriminative leaderboard: only the offline robust batch WLS reference produces a meaningful postfit solution. A controlled force-model/process-noise mismatch split inflates the truth drag/SRP/process-noise force model while estimators keep the nominal compact model. Bounded public precise-reference probes score real ILRS satellite-laser-ranging normal points for LAGEOS-1 and LAGEOS-2 [59, 27], parsed from public daily CRD version-2 files [26] archived by the ILRS/NASA CDDIS data centers [53]. None of these slices is operational flight validation.

The public-data inventory and the per-scenario provenance digests are recorded in S-Table 37.

3.2 Evaluation tasks

SPOT-OD is an evaluation suite. The evaluated tasks are state estimation (acceptance-critical) and method selection (an internal synthetic routing diagnostic; see S-Table 38). Stability labels serve exclusively as computational guardrails for the engineering-failure audit defined below.

A trajectory is numerically divergent when its trajectory-level position RMSE is non-finite or exceeds 10^8 m, or its velocity RMSE exceeds 10^5 m/s; method-selection costs assign a 10^8 m penalty to divergent candidates. These are computational guardrails, not physical stability certificates, so the results add a separate engineering-failure audit at 100 km position RMSE or 100 m/s velocity RMSE.

3.3 State, dynamics, and observations

Each trajectory has a six-dimensional Earth-centered inertial state

$$x_t = [r_x, r_y, r_z, v_x, v_y, v_z]_t, \quad (1)$$

with position in meters and velocity in meters per second. The generator uses a fourth-order Runge–Kutta Cartesian ECI propagator with 20 s steps for 120 steps. Nominal sampling uses altitude 450–950 km, eccentricity 0–0.02, inclination 20–98 degrees. The acceleration model includes two-body gravity, J_2 , drag against a co-rotating exponential atmosphere, optional point-mass Sun/Moon third-body terms, and optional solar-radiation pressure with a cylindrical Earth-shadow test. The reported nominal/stress splits use ballistic coefficient $0.018 \text{ m}^2/\text{kg}$, $\rho_0 = 4.0 \times 10^{-11} \text{ kg/m}^3$, $h_0 = 400 \text{ km}$, $H = 60 \text{ km}$, third-body and SRP enabled, area-to-mass $0.02 \text{ m}^2/\text{kg}$, $C_R = 1.35$, and zero truth process noise. The force model is intentionally compact rather than flight-dynamics complete: it omits high-order geopotential, precession/nutation, empirical atmosphere, maneuvers, and sensor-scheduling uncertainty beyond the specified perturbations.

The measurement model is a station–satellite line-of-sight observation $z_{t,s} = [\rho, \alpha, \epsilon, \dot{\rho}]_{t,s}$ (range, azimuth, elevation, range-rate); a station is visible at elevation ≥ 8 degrees. The nominal configuration uses eight ground stations. Nominal measurement noise is range 30 m, azimuth/elevation 0.02 degrees, range-rate 0.08 m/s, outlier probability 0.01, outlier scale 6, dropout 0.02, with small station/clock bias. The reported stress-test override increases range noise to 70 m, azimuth/elevation noise to 0.06 degrees, range-rate noise to 0.22 m/s, outlier probability to 0.04, and outlier scale to 10. These are protocol facts, not inferred from result tables.

3.4 Splits, seeds, and evaluation units

The main experiment uses seed 42. The dataset contains 160 nominal training, 32 nominal validation, 64 nominal test trajectories, 96 stress-training, 24 stress-validation, and 64 stress-test trajectories. The curriculum uses nominal pretraining, mixed training, and stress fine-tuning; later validation includes a

SatNOGS replay split. All methods are scored on steps 11–119 (109 evaluated steps); nominal and stress aggregates summarise $64 \times 109 = 6976$ evaluated trajectory-steps each. The paired statistical unit is the trajectory-wise position RMSE over the same 109-step window, so Wilcoxon comparisons use $n = 64$ paired trajectory values per comparison. All filters and learned models receive the same simulated measurements, visibility masks, station geometry, and noisy initial state estimates (2500 m position, 6 m/s velocity standard deviation); learned residual methods additionally receive the relevant filter priors.

4 Methods Compared

Estimator abbreviations are defined at first use, with a compact main-text guide in Table 1 and a longer glossary at the start of the supplement. The compared families are classical recursive filters (EKF, UKF, AUKF), predeclared adaptive or structural variants (PUKF, DMC-EKF, DSA-EKF), an offline robust batch WLS OD reference, the bounded RGR residual learners and controls, and KalmanNet-related sanity/transposition diagnostics.

4.1 Classical filtering baselines

The EKF [28], UKF, and AUKF estimate the same six-dimensional state from common range/azimuth/elevation/range-rate observations (UKF unscented-transform parameters $\alpha_{UT} = 0.2, \beta_{UT} = 2, \kappa_{UT} = 0$; AUKF adds innovation consistency R adaptation: rate 0.08, measurement scale bounds 0.4–30, Huber 2.8, NIS soft gate 16). The unscented-transform beta parameter is denoted **beta.UT** in prose and β_{UT} in equations; it is distinct from the drag-scale beta. Classical hyperparameters were selected through a 48-trial tuning budget. We also include an offline *robust batch weighted least squares OD reference*: per trajectory it fits one initial Cartesian state to the full visible arc using the same force/measurement model and a Huber robust objective. This is intentionally non-causal and postfit — a classical OD reference that can use future measurements, not a deployment equivalent comparator. We additionally include a *process noise adaptive UKF (PUKF)* as the symmetric counterpart of the AUKF: it holds R fixed at the nominal vector and inflates the predicted process noise Q multiplicatively when a sliding window estimate of the R -only normalized innovation squared exceeds the chi-square-with-4-channels expectation, on the classical statistical OD intuition that under a dynamics/force model bias the prior model, not the measurement channel, carries the misspecified component [48, 28, 75]. The PUKF rule, thresholds, and decision predicate are fixed in a timestamped rule record.

Ensemble Kalman and sequential Monte Carlo OD/data-assimilation methods are a distinct comparator family rather than a small variant of the tuned EKF/UKF/AUKF guardrail used here; they introduce ensemble-size, stochastic process-parameter, resampling, and sampling-noise design choices, and recent OD work explicitly evaluates EnKF and particle-filter tracking/parameter-estimation variants in that family [76]. To close the ensemble-Gaussian-filter exclusion rather than only cite it, we add one *canonical untuned stochastic perturbed-observation Ensemble Kalman Filter (EnKF)* [16, 6] as a bounded auxiliary comparator: a 64-member ensemble propagated through the same compact two-body, J_2 , and exponential-drag dynamics and updated with the same range/azimuth/elevation/range-rate measurement, process-noise, and azimuth de-weighting model as the EKF/UKF/AUKF baselines, with inflation and localization deliberately left off and a per-trajectory seed fixed in an internally frozen, hash-recorded rule that is an internal pre-evaluation record, not an externally timestamped pre-registration. Because the EnKF shares its process-noise and initial-covariance settings with the UKF and AUKF, the controlled comparison is EnKF versus UKF/AUKF, while the EnKF-versus-EKF gap additionally reflects the EKF’s slightly larger tuned process noise. On the force-mismatch, nominal, and stress slices the untuned EnKF does not displace the classical filters (S-Table 4); in the current rerun its mean observed-step RMSE is 501.5, 390.0, and 1391.0 m versus best non-EnKF means of 431.2 m (EKF), 376.4 m (EKF), and 837.3 m (AUKF), respectively, so the predeclared positive criterion is unmet in all three slices. It is therefore reported as a bounded result for this single untuned EnKF configuration; tuned, inflated, or localized EnKF variants and particle/Gaussian-mixture filters remain outside this comparator set.

4.2 Learned estimator architecture and inputs

Station nodes contain the scaled four-channel measurement, a visibility indicator, Earth-radius-scaled station coordinates, and optional innovation features. Graph-enabled models apply message-passing layers over an implicit fully connected station graph; graph-disabled variants either skip those layers or replace them with equal-depth local per-station layers. Visible embeddings are pooled and passed through a bidirectional GRU over a 12-step window. The training uses hidden dim 192, five graph layers (when enabled), two GRU layers, dropout 0.10, batch 128, lr 10^{-3} , weight decay 10^{-4} , grad-clip 1.0, mixed precision, early-stop patience 10, and a 24/18/12-epoch nominal/mixed/stress curriculum. The full architecture and regularizer weights are in Appendix 14; the supplementary evidence materials supplied for independent inspection provide the complete reproduction records.

Hyperparameter and selection transparency. All compared families expose their fixed settings either in this Methods section, the reproduction checklist, or the supplement: EKF/UKF/AUKF use the common state, dynamics, observation, and noise settings stated above with the UKF and AUKF parameters stated here; PUKF, DMC-EKF, and DSA-EKF use predeclared rule records and the Gauss–Markov hyperparameters listed in the reproduction checklist; prior-guided learned variants share the same inputs, backbone dimensions, curriculum, optimizer settings, bounded-residual scale, gate, context budget, and prior-anchoring penalty stated here and in Section 4.3; and KalmanNet-related checks use their separately scoped native-benchmark, transposition, and validation-grid records. Selection fairness is enforced before held-out scoring: for each held-out slice the candidate family, validation grid, selected setting, primary metric, comparator set, practical floor, seed/split identifiers, and pass/fail rule are fixed before the test population is evaluated, and no method is retuned after seeing held-out outcomes. Validation-tuned choices are linked to validation records rather than test outcomes, and trained-model/evaluation records are included so that every displayed row can be inspected without inferring hidden hyperparameters from the result tables.

4.3 Method families

Classical filters are EKF, UKF, AUKF, and the predeclared process noise adaptive UKF (PUKF). **The classical postfit OD reference** is robust batch WLS, reported separately because this *postfit classical OD reference* is offline rather than causal.

The published learned OD baseline is the KalmanNet release [64], exercised on its own linear canonical benchmark as the upstream architecture sanity reproduction (S-Table 36). The in-house KalmanNet-style learned-gain comparator keeps the EKF prior and flow and replaces the analytic gain with a bounded learned gain correction driven by normalized innovations and shared features; its gain scale and correction clip are selected by held-out validation loss over a small predeclared grid with no test-set information. It is a load-bearing in-house boundary comparator, not an official KalmanNet reproduction or exhaustive design-space test; the documented KalmanNet transposition is retained separately as a feasibility/design-gap diagnostic, not architecture-level refutations or learned-claim evidence.

Prior guided learned variants (flagship RGR-GF plus No-Graph Residual, Learned Noise Adaptive, IDP-RGR-GF, OI-RGR-GF) share a common backbone: a bounded residual $\hat{x} = \hat{x}_{\text{prior}} + g \odot (b s \tanh(r_\theta))$ with $s = 0.03$, $g \in (0, 1)^6$, $b \in (0, 1)$, plus a $0.05 \|\hat{x} - \hat{x}_{\text{prior}}\|^2$ prior anchoring penalty; the gate g and budget b form a per-component learned trust region around the fused prior, whose variance-minimising operating point collapses toward the prior on the majority of steps where the visible measurements expose little prior-unexplained residual, so any “competitive with classical” behaviour is in substantial part inherited from the classical prior bank by construction; the **EKF residual learner** is a learned residual control (not a KalmanNet reproduction), retained only as an ablation row to bound the EKF only residual variant; it diverges substantially from EKF on the all-step propagation reference (S-Table 40) and is excluded from the claim audit’s pairwise comparisons because it does not share the bounded residual contract of the RGR-GF family. **Input matched message passing controls** are RGR-noMP (skip control bypassing cross station aggregation) and RGR-local (stricter capacity matched control replacing the graph stack with equal depth, equal width local per-station layers). The matched budget ablations share splits, training stages, hyperparameter budget, and evaluation window, not exactly equal parameter counts or inputs. An unconstrained **Pure GNN** state predictor (no filter prior) is retained only as a supplementary sanity-check

row (S-Table 69); it diverges by orders of magnitude on every slice and is not used in any audit comparison.

4.4 Observability-geometry features and guarded selection

Per step, SPOT-OD computes a local unweighted measurement-geometry Gramian from finite-difference Jacobians of the normalized measurement model with respect to scaled state, $\mathcal{I}_t = \sum_{s \in \mathcal{V}_t} H_{t,s}^\top H_{t,s}$. The summary uses clipped log trace, log pseudo-determinant, log condition number, rank fraction, visible-station fraction, and minimum-eigenvalue features (causal: computed from the filter prior, not truth). The guarded selector is a separate trajectory-level cost predictor over visibility, public/stress flags, innovation energy, prior disagreement, and these summaries; it is an internal evaluation safeguard, not a proof of operational autonomy.

4.5 Statistical protocol

Endpoint terminology (defined once, used throughout). *This study is exploratory*¹: the commitments are internal predeclarations supplied with the submission, with timestamped records where the record format supports them, not external trial-registry preregistration or confirmatory validation. *Predeclared* means a metric, threshold, or decision rule written down before the run that produced its result. The $K = 16$ frozen-rule extension has a confidential RFC3161 timestamp of its rule hash (hash disclosed, not content); the timestamp confirms rule-before-evaluation ordering for that draw but is not public preregistration and does not upgrade the observed-step endpoint itself from post-hoc to externally predeclared. Evidentiary control comes from retained internal rule records, fixed seeds/splits, and digest-linked evaluation records, which make post-hoc rule substitution inspectable under internal frozen-rule consistency but do not provide confirmatory external validation.

Endpoint timeline (summarized; full details in supplement): Observed-step RMSE was selected post hoc on the training cohort before $K = 32$ generation. $K = 8$ internal record froze the selected endpoint on a seed-disjoint draw (no external timestamp). $K = 16$ was dependent, informed $K = 32$ rule handling, and is non-evidentiary. $K = 32$ decision rule was fixed after those records but before $K = 32$ generation (internal record, not external preregistration). $K = 96$ stress-only rule predates its evaluation by internal timestamp evidence (gap 2026-05-25T13:06:32Z to 13:12:43Z confirms temporal ordering; not external preregistration). $K = 8/K = 32/K = 96$ are internal consistency/replication checks under the selected endpoint, not external confirmatory evidence or endpoint-family validation. $K = 16$ (seed 1840000) has confidential RFC3161 hash timestamp, stronger than internal timestamp for that draw but not public preregistration. Protocol-subset sufficiency audit (S-Table 15) shows omitting any ingredient would have admitted misleading positives.

Other internal commitments include the larger independent $K = 32$ endpoint replication under its frozen decision rule, the stress-only $K = 96$ floor-power-scale replication, the DBAR decision rule, and the network-consistent dense-tracking probe’s primary metric. *Protocol-fixed endpoints* (definitions fixed by the evaluation protocol, not chosen after inspecting results) are the controlled force-model-mismatch split and the multi-revolution SGP4-reference benchmark. *All-step trajectory position RMSE is retained only as a propagation-dominated reference* (roughly four-fifths of evaluated steps have zero visible stations); it is reported symmetrically as a divergence/failure check, never as a headline endpoint. For the compact simulator audit, we fix before interpreting any ordering a *practical significance floor* of 3% of the observed-step position RMSE: a learned versus classical difference smaller than this floor is treated as *below-floor* regardless of statistical detectability. This is a mission-agnostic audit threshold for this compact simulator, not a universal OD mission requirement; a mission-specific OD campaign would replace it with its own acceptance threshold. By this floor the nominal synthetic split is non-discriminative (full learned versus classical spread $\approx 1\text{--}3\%$ on ~ 385 m), so its single-seed tables are illustrative only and the discriminative conclusions rest on splits where the observed effect exceeds the floor (measurement-noise stress, controlled force-model mismatch, the $K = 8$ endpoint-fixation support draw, the larger simulator-bound $K = 32$ endpoint replication, and the stress-only $K = 96$ replication).

¹In statistical methodology, *exploratory* analysis selects endpoints or hypotheses after inspecting data patterns, while *confirmatory* analysis tests pre-registered hypotheses on held-out data. This study’s endpoint timeline is exploratory with internal frozen-rule replications ($K = 32$ and $K = 96$ draws), not externally pre-registered confirmatory validation.

Significance uses paired trajectory-wise position RMSE; the difference is baseline minus candidate (positive favours the candidate). The one-sided Wilcoxon signed-rank alternative is “candidate better” [85]. Mean-gain intervals are percentile paired-bootstrap intervals with shared trajectory indices, at least 3000 bootstrap resamples, and 95% confidence [15]; endpoint-specific tables state the exact count, with structural-channel tables using 5000 paired resamples. These intervals treat the trajectory as the exchangeable resampling unit within a declared slice; they preserve within-trajectory temporal dependence, including within-pass residual correlation, but do not turn consecutive visible updates into independent units. The limitation is that trajectory-pair resampling does not cover dependence from repeated trajectory identities across trained seeds, simulator-design uncertainty, selection uncertainty, possible correlation among similarly generated passes, or transfer beyond the finite simulator and public-probe populations. The low visible-step fraction (roughly 20%) limits but does not eliminate within-pass correlation as an uncertainty concern. Seed-bootstrap intervals are likewise descriptive variability summaries over the retained seed ensemble, not independent scenario-population confidence intervals. Decisions are CI-based at the predeclared 95% level (the two-sided upper-bound rule, stricter than a one-sided $\alpha = 0.05$ Wilcoxon). Every p -value is a transparency diagnostic, not a confirmatory test. No familywise multiple-comparison correction is claimed for any individual table; for completeness S-Table 39 reports Holm familywise- and Benjamini–Hochberg false-discovery-rate-adjusted values over the entire family of displayed pairwise Wilcoxon p -values [24, 4]. The Holm/BH adjusted p -values are transparency diagnostics only; the CI-based predeclared decision rule is unchanged and the qualitative ordering and the disciplined negative remain unchanged under both adjustments.

5 Results

Load-bearing evidence roadmap. The central C1 mechanism finding is grounded in Table 5, Figure 1, and Table 7: AUKF inflates effective R under compact force-model mismatch (mean 3.32), and the drag-scale cascade separates EKF linearisation, sparse-geometry observability, and candidate-inertia failure modes, with median trajectory-mean $\hat{\beta} = 1.0000$ despite 9.6 km arc-mean drag separation at enriched geometry. The secondary learned-estimator result is a bounded negative supported by Table 4 plus the stress-only and all-scenario internal replications in S-Tables 17 and 19: no evaluated learned construction beats the per-scenario best tuned classical reference on the primary observed-step endpoint. Those added $K = 96$ records are internal consistency checks, not a stronger preregistration tier. The DBAR withdrawal is reported in Table 9: 450-realization characterization is statistically indistinguishable from the 81.3% no-information baseline. Structural/adaptive channels are judged only by predeclared pass/fail rules, and public catalog/SP3/ILRS slices are used only for provenance, measurement-pipeline, or mechanism-transfer diagnostics.

5.1 Primary observed-step view and propagation reference

The central $K = 32$ observed-step anchor and the observed-step/all-step endpoint-choice sensitivity audit both return no learned positive. The *primary observed-step state-estimation view* uses the larger seed-disjoint frozen-rule endpoint replication under the selected observed-step endpoint hierarchy as its central evidence anchor (Table 4); the full post-hoc endpoint timeline is defined in Section 4.5. The replication evaluates 24 trajectories per realization and reports no learned positives: nominal is best with EKF and an RGR-GF-minus-best-classical gap of +5.4 m (CI $[-15.7, 31.3]$ m), stress is best with AUKF and a gap of +48.8 m (CI $[+17.7, +88.8]$ m), and controlled force mismatch is best with EKF and a gap of +16.8 m (CI $[+6.1, +27.8]$ m). This larger replication is simulator-bound per the Results-scope convention and Table 2 above.

The nominal split is reported for traceability and architecture-selection debugging only, not for pass/fail conclusions: by the predeclared 3% floor and the later power audit, its learned-versus-classical spread is at or below practical significance and underpowered for decision use. Retaining underpowered nominal rows avoids suppressing full-scenario evidence that supports architecture-selection debugging and future meta-analysis, but nominal rows must not be used to draw discriminative learned-superiority conclusions. The stress and controlled-mismatch rows carry discriminative interpretations for their observed degradations, and the predeclared structural/noise-adaptive channels and public-provenance probes carry their own bounded decision interpretations.

Here “independent” means an independently seeded trajectory population and measurement-noise draw per realization, not independent time steps within one trajectory. A post-hoc analytical power check using

Table 3: One-page findings summary for the three load-bearing outcomes. “Decision” states what the evidence supports at submission time; “Scope” states what it does not support.

Finding	Evidence	Key numbers	Decision	Scope
AUKF force-mismatch mechanism	Table 5, Figure 1, Table 7, S-Table 42	Mean effective- R scale 3.31; median R -only NIS 4.73 (AUKF) vs 1.60 (EKF) and 1.83 (UKF); EKF–AUKF paired mean -58.1 m on the compact mismatch slice	Supported as the primary compact-model mechanism diagnostic	Not an operational EKF>AUKF rule; long-arc higher-fidelity slices show regime dependence
Exploratory learned negative on the observed-step endpoint	Table 4, S-Table 17, S-Table 19	No learned positives under the per-scenario best-tuned-classical rule at $K = 32$; the outcome is no learned positives in the stress-only $K = 96$ check and no learned positives in the all-scenario $K = 96$ seed-disjoint replication	Supported as a secondary internal frozen-rule negative under the selected observed-step endpoint	Not external preregistration, not a claim about all learned OD systems, and not operational POD validation
Inspection, reproducibility, and support-only public probes	Table 9, Table 6, Section 6, Section 10	DBAR characterization 81.8% vs 81.3% no-information baseline; source-tree active-regeneration report records 10/10 passes, 0 mismatches, and 0 blockers; archive-extracted tier reruns 10 public CRD/SP3 arcs with exact submitted-table-text recovery	Supported as an inspectable compact-screening record with explicit withdrawals, validation artifacts, and support-only measurement-pipeline checks	Covers inspection readiness and measurement-pipeline support only; higher-fidelity and public-reference checks remain bounded diagnostics rather than operational validation.

the realization-level paired-gap standard deviations is reported only at the predeclared 3% floor effect, not at the realized gap, because power computed at an observed effect is a deterministic function of the realized p -value and adds nothing beyond the interval already shown: it gives limited sensitivity to an exactly floor-sized stress effect (3% of the AUKF mean, 27.5 m; approximate one-sided $\alpha = 0.05$ power ≈ 0.42), and the realized stress gap of 48.8 m exceeds that floor. These power figures use a one-sided t -approximation at $\alpha = 0.05$, whereas the decision predicate is the stricter two-sided 95% CI upper-bound rule (effective one-sided $\alpha = 0.025$), so the realized floor power is somewhat below 0.42 and the K required for about 0.80 power somewhat above the figure below. With the realized stress-row paired-gap SD of 106.2 m, the same one-sided t approximation would require $K \approx 94$ independent realizations for about 0.80 power at the 27.5 m floor. The completed stress-only $K = 96$ internal replication (S-Table 17) exceeds that design check for approximately 0.80 power at the 27.5 m floor and again finds AUKF best on observed-step RMSE: AUKF 923.5 m versus RGR-GF 955.5 m, with RGR-GF-minus-AUKF +31.9 m (CI [+12.4, +52.4] m). The mean gap exceeds the 27.5 m floor while the CI lower bound of +12.4 m lies below that floor, so the floor-clearing criterion is not met; this does not alter the predeclared predicate because RGR-GF is not best and the paired interval is not strictly negative. It is a $K=96$ internal replication under an already selected endpoint, not external preregistration or public-reference validation. The controlled-mismatch row is stronger for the floor itself (3% of the EKF mean, 14.2 m; power ≈ 0.79), and its realized gap of 16.8 m also exceeds that floor. Thus the endpoint anchor plus the powered stress-only check support the discriminative negatives, while the nominal row remains underpowered for positive claims. An additional independently seeded internal prospective replication (base seed 1160000, $K = 32$ realizations per scenario, 24 trajectories per realization, freeze record written before any evaluation) is reported in S-Table 18. The replication preserves the bounded

Table 4: Main load-bearing endpoint result: larger seed-disjoint frozen-rule observed-step replication under the $K = 8$ -established endpoint hierarchy. Positive RGR-GF-minus-best-classical gaps mean the learned estimator is worse than the best tuned classical reference. Nominal is retained for traceability only and is not a discriminative endpoint under the predeclared floor and power audit; its CI direction should not be interpreted.

Scenario	Best classical obs. RMSE [m]	RGR-GF obs. RMSE [m]	RGR-GF–best classical [m] (95% CI)	Interpretation
<i>Nominal (traceability only)[†]</i>	EKF 402.0	407.4	+5.4 [−15.7, +31.3]	Traceability only; non-discriminative; do not interpret CI direction
Measurement-noise stress	AUKF 915.8	964.6	+48.8 [−17.7, +88.8]	No learned positive
Controlled force-model mismatch	EKF 474.3	491.1	+16.8 [−6.1, +27.8]	No learned positive

$K = 32$ independent realizations per scenario, 24 trajectories per realization. The daggered nominal row is traceability-only; its CI direction must not be interpreted. The rule was frozen after the $K = 8$ endpoint-fixation results and the non-evidentiary $K = 16$ strict-prefix extension were available, but before any $K = 32$ realization was generated or evaluated; all-step RMSE is propagation-dominated and retained in the supplement. Best classical is formalized per scenario as the lowest held-out mean observed-step RMSE among tuned EKF, fixed-noise UKF, and AUKF under the frozen endpoint hierarchy; realized labels are Nominal=EKF, Stress=AUKF, Mismatch=EKF. The stress row is discriminative for the observed degradation, but this table alone is not powered to exclude every just-at-floor stress effect: using the realized stress-row paired-gap SD, the 27.5 m floor would require approximately $K = 94$ for 0.80 one-sided $\alpha = 0.05$ power. A separate stress-only $K = 96$ internal replication in S-Table 17 exceeds that design check and preserves the negative.

negative and no-learned-positive outcome across all three scenarios: nominal is best with EKF (388.46 m), RGR-GF 391.71 m, gap +3.25 m (CI [−7.64, +18.24] m; nominal remains non-discriminative and is retained for traceability only); measurement-noise stress is best with AUKF (958.41 m), RGR-GF 999.61 m, gap +41.21 m (CI [+10.04, +74.97] m); controlled force-model mismatch is best with EKF (465.12 m), RGR-GF 487.57 m, gap +22.45 m (CI [+14.91, +29.51] m). No scenario returns a learned positive under the frozen predicate. This record is an additional internal evidence check, not external preregistration or operational validation. A further seed-disjoint internal replication (base seed 1630000, 24 trajectories per realization, freeze record written before any evaluation; S-Table 19) carries all three scenarios to the powered $K = 96$ sample size, whereas the earlier $K = 96$ check covered only the stress split: nominal is again best with EKF (402.20 m), RGR-GF gap +3.77 m (CI [−9.63, +20.14] m; non-discriminative, retained for traceability only), measurement-noise stress is best with AUKF (948.49 m), RGR-GF 981.31 m, gap +32.83 m (CI [+8.50, +61.52] m), and controlled force-model mismatch is best with EKF (449.35 m), RGR-GF 470.47 m, gap +21.12 m (CI [+16.44, +25.51] m). No scenario returns a learned positive at the powered all-scenario sample size, so the bounded no-learned-positive outcome holds at $K = 96$ across every scenario and a fresh seed; this too is an internal evidence check, not external preregistration or operational validation.

The 15-seed cohort and the endpoint-fixation records are supporting evidence, not the central anchor. The 15-seed observed-step stress gain of RGR-GF over fixed-noise UKF is 768.32 m (95% seed bootstrap CI [714.85, 821.04] m, all 15 seeds favouring RGR-GF), but versus the tuned AUKF the same stress comparison is −199.18 m (no seed favouring RGR-GF), the nominal-split RGR-GF–AUKF gain is −73.07 m, and the small all-step nominal-split UKF margin does not survive on the primary metric (−78.33 m). The 15-seed record therefore supports only a *ukf-specific stress-gain claim* bounded to *fixed-noise UKF under the documented stress split*, with no upgrade against adaptive filtering. Observed-step RMSE is the physically motivated estimator-skill metric (zero-visible steps are open-loop propagation), and all-step aggregates (S-Table 40) are retained only as a propagation-dominated reference. These outcomes separate along the axis of the scenario shift: a measurement-noise shift calls for the gain re-weighting an innovation-adaptive AUKF already supplies, so the learned margin over the fixed-noise UKF is chiefly recovered noise-adaptation the AUKF represents, whereas a force-model-mismatch shift calls for a structural dynamics correction whose signal is neither observable from sparse line-of-sight updates nor representable by a residual anchored to the compact prior, so the learned residual never beats the causal EKF.

Endpoint-fixation and propagation context. The disjoint endpoint-fixation draw (S-Table 22) records the transition from the earlier all-step metric to the observed-step endpoint and supports the same no-learned-positive ordering as the larger frozen-rule replication. An endpoint-choice sensitivity audit (S-Table 23) re-reads the retained endpoint-support and larger-replication records under both observed-step and all-step position RMSE and finds no learned positive under either natural endpoint choice. The single-seed measurement-informed scoring table (S-Table 30) is illustrative only, and the all-step propagation reference

(S-Table 40) shows *AUKF is the strongest displayed aggregate position-RMSE method* (6167.44 m nominal / 6800.54 m stress; EKF 6170.19 / 6811.25, UKF 6191.37 / 7173.23, RGR-GF 6169.11 / 6808.98). About four-fifths of evaluated steps have zero visible stations, so the all-step view is a propagation snapshot rather than a learned versus classical comparison.

The offline robust batch WLS reference is deliberately separated from causal recursive filtering because it uses the full visible arc to refit the initial state. Its nominal observed-step RMSE (152.61 m in the supplement) is far below the recursive filters, showing that the strongest architecture for this slice is an offline OD fit when latency and future measurements are allowed. This explains why batch/offline guardrails are required before any causal learned estimator is treated as a candidate for operations-facing validation.

5.2 Densified-visibility control and network-consistent dense-tracking probe

After correcting the observation-network inconsistency, the dense-tracking probe still finds no learned estimator beating the best classical filter on its predeclared metric. The dense-tracking probe is a bounded measurement-geometry/observability diagnostic that isolates whether the sparse-geometry learned-negative behavior persists under measurement-update-dominant conditions; it is not a primary discriminative endpoint or headline learned-negative claim. An earlier uncorrected densified-visibility control (S-Table 62) exposes a *measurement-conditioning artifact*: every estimator exhibited multi-thousand-kilometre median errors and a near-identical gross-failure rate, the signature of an estimator/truth observation-network inconsistency (filters resolved the nominal eight-station network while truth was synthesised against the twenty-station densified network). We therefore explicitly withdraw the earlier reading and draw the dense-tracking conclusion only from the corrected, network-consistent probe.

The corrected *network-consistent dense-tracking probe* (S-Table 31) enforces a perfect shared model (identical dynamics with third-body and SRP enabled, identical measurement model, identical dense twenty-station network on both sides), removing the observation-network inconsistency that drove the uncorrected control; every estimator additionally applies the standard capped $1/\cos^2(\text{elevation})$ topocentric-azimuth de-weighting, applied identically and inherited by the learned residual through its classical priors so it cannot manufacture a learned advantage. The *primary metric is predeclared* (fixed before running): unlike the sparse synthetic primary endpoint, this dense, high-visibility, network-consistent setting uses pooled all-step trajectory position RMSE over non-divergent trajectories as a stress test for propagation/control over the full dense arc, with observed-step pooled RMSE reported *symmetrically* as the secondary, plus heavy-tail-robust medians and a 100 km gross-failure rate. The catastrophic divergence collapses to a small minority and classical filters reach sub-kilometre-to-kilometre state error; the bounded result persists on the predeclared metric, the secondary, and the heavy-tail-robust median (the best classical filter is best on all three; no learned estimator beats it). A fixed, previously trained learned estimator is evaluated without any per-realization refitting.

Tail-conditioned audit. The predeclared paired learned-minus-best-classical comparison on S-Table 31 is dominated by a small number of megametre-scale tails. S-Table 32 restricts the paired comparison to the jointly engineering-adequate subset of 111/120 trajectories (every estimator below the 100 km physical-adequacy threshold on a given trajectory) so the mean cannot be dominated by such tails, and reports per-method gross-failure counts (EKF 0, UKF 8, AUKF 4, RGR-GF 6 of 120) so the per-realization tail structure is auditable. On the tail-conditioned engineering-adequate subset the learned versus classical bounded result is preserved on the predeclared pooled and paired-mean views in both the all-step and observed-step metrics; the paired *median* difference is slightly negative (RGR-GF lower median error) on the same subset, so the bounded result on the predeclared mean is consistent with RGR-GF having a typical-case advantage offset by a heavier mid-tail than the best classical reference.

5.3 Controlled force-model mismatch and mechanism

The controlled compact mismatch quantifies an OD instance of AUKF R -inflation damping, with EKF best among the recursive filters and no learned residual beating EKF or AUKF on this slice. The long-arc higher-fidelity slice (Section 5.9, S-Table 29) is a model-scope warning rather than a transfer rule: it contains a supplementary EKF/AUKF direction-only stress observation (EKF-minus-AUKF paired interval

[+17.5, +637.7] m, auxiliary direction-only stress observation, not a transfer rule or estimator positive), with details retained in the supplement. The compact- J_2 finding reported in this subsection is a *bounded counterexample* to that warning, presented as a mechanism observation about the covariance-matching family on a deliberately compact dynamics model and not as an operational EKF-over-AUKF prescription; reading both findings together, the long-arc stress observation motivates confirmation before any compact-model classical-filter prescription is transferred.

The controlled force-model/process-noise mismatch inflates ballistic coefficient (0.045 vs 0.018 m²/kg), process-noise std (0.45 vs 0.0), drag density (7.0×10^{-11} vs 4.0×10^{-11}), SRP area-to-mass (0.06 vs 0.02), and C_R (1.60 vs 1.35) on the truth side, while every estimator and the WLS reference keep the nominal compact model. The full per-method observed/all-step table is in Appendix 15 (S-Table 41); the discriminative paired evidence is the mechanism diagnostic below. Over the 38-trajectory population the deterministic classical filters pair (S-Table 42) as: causal EKF beats tuned AUKF by 58.1 m mean observed-step (95% paired bootstrap CI [22.2, 100.0] m, $p = 0.015$ descriptive, 65.8% of trajectories) and beats fixed-noise UKF by 26.1 m (CI [3.4, 54.5] m, $p = 0.041$); fixed-noise UKF in turn beats AUKF (32.0 m, CI [0.9, 67.4] m). These nominal-level paired intervals exclude zero, but the corresponding familywise-adjusted p -values (Holm/BH; S-Table 39) do not reach significance, so the compact- J_2 ordering is reported as a predeclared CI/direction reading bounded to this slice—consistent with the higher-fidelity zero-crossing and the long-arc EKF/AUKF reversal disclosed below—rather than a confirmatory test. Comparing each learned construction against the per-scenario *minimum*-of-three classical filter is a deliberately stringent bar that, if anything, inflates the classical-side gap magnitudes, so the learned negatives are conservative. The pooled per-method observed-step means in S-Table 41 are computed over the 48 processed trajectories, whereas these paired gains use per-trajectory RMSE over the 38 trajectories with a finite shared-mask observed-step RMSE; a paired mean of differences is therefore not the difference of pooled means, and the two are not expected to coincide. Across all 15 trained seeds the learned residual never beats EKF on this slice (0/15 seeds; seed-mean gap −99.9 m, CI [−157.4, −51.8] m) and does not beat AUKF either (CI [−82.5, 25.6] m, 9/15 seeds favouring RGR-GF with the interval crossing zero and the mean against).

Figure 1, Table 5, and S-Table 24 give the paper’s *compact-model mechanism diagnostic*: a deterministic, exactly reproducible decomposition (it reproduces the reference AUKF estimate to 0.00 m). The visible-update innovations are dominated by an unmodelled dynamics bias, yet the AUKF inflates its effective R scale (mean 3.32, from 2.89) and damps corrections, so its mean state-update norm collapses to about 466 m; the cross-filter R -only median NIS grows from 1.60 (EKF) and 1.83 (UKF) to 4.73 (AUKF). The damping is the direct gain consequence of the inflation: the Kalman gain scales as $K \propto P^- H^\top (H P^- H^\top + sR)^{-1}$, so scaling the effective measurement noise by s (here mean 3.32) monotonically shrinks the gain and the correction norm. Under white measurement noise this attenuation is appropriate; under a persistent, serially correlated dynamics bias it attenuates precisely the updates that would track the bias. This is the *force-model mismatch mechanism diagnostic*: the adaptive filter *misreads dynamics-bias innovations as measurement-noise* evidence and damps the updates that would track the bias, a known covariance-matching/adaptive-filtering failure mode [48, 65, 28, 75]. The NIS and effective- R -scale values are single-population deterministic descriptors of the controlled-mismatch population; the cross-seed robustness of their *direction*, not these point magnitudes, is what the R -inflation cap control (S-Table 28) and the 38-trajectory paired CIs evidence.

As an additional targeted mechanism-control, S-Table 28 runs AUKF with effective R scale capped at $2 \times$ nominal (AUKF-Rcap) across all 48 force-mismatch trajectories. AUKF-Rcap inherits all standard AUKF parameters unchanged and differs only in the maximum effective R -scale ceiling, which is reduced from $30 \times$ to $2 \times$. All 48 trajectories are processed; the paired denominator is 38 because 10 trajectories have zero visible measurements after the scoring window starts at step 11, making observed-step trajectory RMSE undefined for all comparators under the shared visibility/scoring mask—these are not AUKF-Rcap- or standard-AUKF-specific nonfinite failures. All paired CIs and win rates are computed over the 38 shared finite observed-step trajectories. Capping the inflation reduces the mean effective R scale from 3.32 to 1.09 and improves the pooled observed-step RMSE by 41.7 m relative to standard AUKF (paired gap −23.6 m, 95% CI [−42.3, −6.2] m; 63.2% win rate over 38 paired finite trajectories, above the 50% equal-direction reference; this is an additional targeted mechanism-control outside the original frozen learned-estimator claim audit and not a predeclared positive criterion). AUKF-Rcap remains strictly worse than EKF (+34.5 m, CI [+9.8, +63.9] m), supporting the interpretation that R -inflation damping is a key but not sole driver of the AUKF degradation under compact-model dynamics bias.

AUKF R-inflation mechanism under controlled force-model mismatch

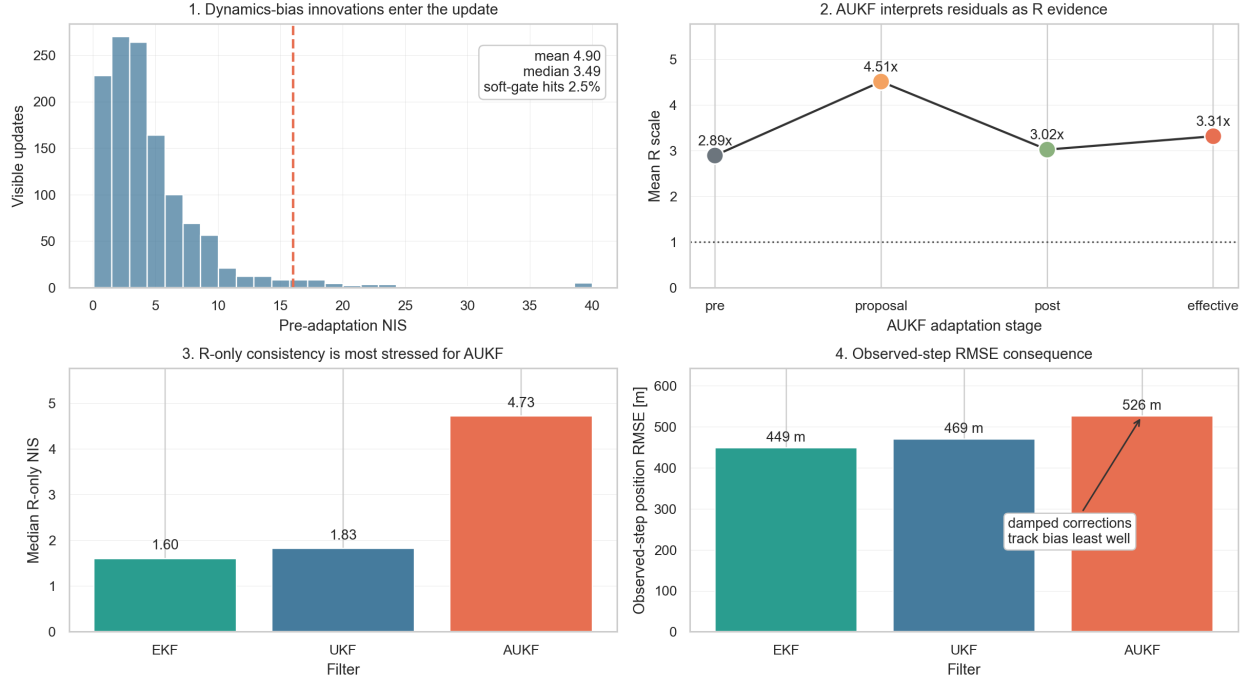


Figure 1: Main-text mechanism view of the controlled force-model-mismatch AUKF diagnostic. Dynamics-bias innovations raise the adaptation signal, and the AUKF inflates effective R (mean scale 3.32, from 2.89) while EKF and UKF remain near nominal scale (mean 1.00, 0.99 respectively). Median R -only NIS grows from 1.60 (EKF) and 1.83 (UKF) to 4.73 (AUKF), and AUKF shows worse observed-step position RMSE than the fixed-noise references on this force-mismatch slice. The figure is derived from the same deterministic compact-model force-mismatch diagnostic summarized in Table 5 and S-Table 24, not from a new run.

5.4 Higher-fidelity force-mismatch (second-fidelity scoping)

Higher-fidelity force-mismatch slices preserve the R -only NIS stress signature but do not support transferring the compact EKF-over-AUKF ordering. S-Table 25 repeats the controlled force-mismatch protocol with the truth-side dynamics extended above the compact-J2 ceiling: the truth propagator adds the J3 and J4 zonal geopotential terms (analytic gradient of the standard zonal expansion) and luni-solar third-body acceleration (analytic low-precision ephemerides at absolute UTC epoch), while every estimator still keeps the nominal compact two-body+ J_2 +drag model. On this richer dynamics mismatch the cross-filter R -only NIS diagnostic of S-Table 24 continues to fire (AUKF median R -only NIS 1.53 vs. EKF 1.37 and UKF 1.47, AUKF p90 5.70), so the compact-model mechanism reproduces qualitatively. The magnitude of the AUKF damping is, however, no longer load-bearing in the EKF/AUKF comparison: the tuned AUKF is the best classical filter (393.3 m mean observed-step RMSE), the causal EKF is second (420.9 m), and the EKF–AUKF paired-bootstrap CI spans zero ($[-152.5, +243.8]$ m). The predeclared PUKF again fails its positive criterion (significantly worse than AUKF by +126.4 m, CI $[+14.9, +296.3]$ m; Section 5.5, S-Table 33). The AUKF caveat from S-Table 24 is therefore a *compact-model* observation: above the compact-J2 ceiling the mechanism continues to fire but the EKF/AUKF ordering is not flipped, so we downscope the compact-model diagnostic. A yet richer fidelity slice (S-Table 35, J5+J6 zonal terms plus a one-sided diurnal-bulge density modulation, $\alpha = 0.30$) confirms the same downscoping: the cross-filter R -only NIS again flags AUKF as the most stressed (median 1.67 vs. EKF 1.42, UKF 1.60) but the EKF/AUKF ordering does not flip (EKF–AUKF mean -21.3 m, CI $[-160.4, +87.1]$ m spans zero), and the best classical filter is the fixed-noise UKF rather than EKF.

Table 5: Compact-model AUKF mechanism diagnostic on the controlled force-model mismatch. The deterministic decomposition reproduces the reference AUKF estimate to 0.00 m and shows the known covariance-matching failure mode in OD form: dynamics-bias innovations are treated as measurement-noise evidence, inflating effective R and damping the updates needed to track the biased truth. Values are aggregated over the full controlled force-mismatch mechanism population (48 processed trajectories); visible-update counts and update-norm statistics are computed over the visible AUKF update records in that population.

Diagnostic quantity	Value
Visible measurement updates	1238
Pre-adaptation NIS (mean / median / p90)	4.90 / 3.49 / 9.16
Effective R scale (pre / proposal / post / effective)	2.891 / 4.513 / 3.021 / 3.315
Mean state-update norm [m]	466.31
R -only median NIS (EKF / UKF / AUKF)	1.60 / 1.83 / 4.73
AUKF R -only p90 NIS	20.38
Observed-step RMSE (EKF / UKF / AUKF) [m]	448.79 / 469.41 / 526.39
AUKF estimate max position difference vs reference [m]	0.00

5.5 Process Noise Adaptive UKF (predeclared)

PUKF fails its predeclared positive criterion on both the controlled compact and higher-fidelity force-mismatch slices. S-Table 33 reports the *predeclared symmetric Q-adaptive UKF (PUKF)* on the same 38-trajectory controlled force-mismatch population. The rule, thresholds, and decision predicate (PUKF is a positive contribution iff its mean observed-step RMSE is strictly lowest and the paired-bootstrap CI versus every comparator is strictly below zero) were predeclared in a timestamped rule record before the evaluation; no rule retuning is allowed. Outcome: PUKF inflates the predicted process covariance heavily (mean Q -scale 5.9) but does *not* satisfy the predeclared positive criterion — its mean observed-step RMSE is worse than the causal EKF (+42.6 m, CI [+15.4, +74.0] m), and the CI versus tuned UKF spans zero (+16.5 m, CI [-7.0, +38.8] m). On the higher-fidelity slice (S-Table 25) the predeclared positive criterion likewise fails (PUKF significantly worse than AUKF, CI [+14.9, +296.3] m). A post-hoc 16-point validation-grid stress for PUKF on that higher-fidelity slice (S-Table 34) also does not rescue the comparator: the validation-selected PUKF remains worse than AUKF on the retained held-out population (selected PUKF 527.5 m vs. AUKF 393.3 m; paired gap +134.2 m, CI [-9.5, +382.2] m over $n = 38$ finite paired trajectories). The drag-scale cascade supplement also records a separate 12-hour observability-grid case in which PUKF overflows on the unaugmented UKF-family stress slice; that is a scope warning for long-arc sigma-point conditioning, not part of the predeclared 40-minute PUKF decision. Because this CI spans zero and is wide, the sensitivity supports only the qualitative conclusion that this bounded validation grid does not rescue PUKF; it does not establish a tuning-sensitive ordering. This sensitivity is reported for tuning comparability only and does not replace the predeclared PUKF evidence. PUKF is therefore a *predeclared bounded negative* at both fidelities, with an added post-hoc comparability stress, and the remaining diagnostic focus is structural (Sections 5.6, 5.7) rather than statistical noise adaptation alone.

5.6 Predeclared Cartesian structural-channel response (DMC-EKF)

The compact DMC-EKF structural-channel check does not beat UKF on the higher-fidelity force-mismatch slice. Stacey and D’Amico’s 2021 adaptive, dynamically constrained process-noise OD work motivates structural process-noise responses to dynamics/force-model bias; the DMC/DSA implementations here are compact, simplified structural-channel checks motivated by that lineage, not faithful reproductions of the full adaptive dynamically constrained formulation and not a rejection of that broader prescription [75].

Section 5.5 reports a predeclared symmetric Q -adaptive UKF (PUKF) as the statistical counterpart of the innovation-adaptive AUKF; both PUKF and AUKF adapt the noise distribution on the R or Q side without changing the deterministic flow. Classical statistical OD prescribes a complementary *structural* response when the dominant residual is a dynamics/force-model bias: augment the state with a low-frequency empirical-

acceleration channel and estimate it jointly with position and velocity, the dynamic model compensation (DMC) recipe [77, 86, 75]. S-Table 26 reports the predeclared DMC-EKF on the higher-fidelity force-mismatch slice (truth-side J2..J6 zonal geopotential plus luni-solar third-body plus one-sided diurnal-bulge atmospheric-density modulation; estimators on the compact two-body+ J_2 +drag deterministic flow). The augmented state appends a per-axis first-order Gauss–Markov empirical-acceleration channel w with predeclared steady-state standard deviation $\sigma_w = 1.0 \times 10^{-6} \text{ m/s}^2$ and decorrelation time $\tau = 300 \text{ s}$; the deterministic flow is shared with the EKF baseline. The predeclared positive criterion (DMC-EKF strictly lowest in mean observed-step RMSE, strictly negative paired-bootstrap CI versus the best non-DMC comparator, and the 3% practical significance floor exceeded) was committed in a timestamped rule record, and no rule retuning is allowed. Outcome: the predeclared criterion is *not* satisfied — UKF gives the lowest mean (276.13 m), DMC-EKF tracks EKF to numerical precision (paired mean difference $7.8 \times 10^{-6} \text{ m}$), and the DMC-EKF–UKF paired-bootstrap CI spans zero (+2.9 m vs. UKF, $[-3.8, +8.1] \text{ m}$). The cause is structural: under sparse-visibility geometry the empirical-acceleration channel is essentially observable only through position/velocity cross-covariance accumulated between visible updates, and the median maximum estimated $|w|$ remains near $1 \times 10^{-10} \text{ m/s}^2$, far below the unmodeled-physics magnitude. The near-zero DMC-EKF–EKF deviation is consistent with the sparse-visibility observability bound rather than with a process-covariance-augmentation implementation defect: a separate favorable-geometry sanity diagnostic in S-Table 3 shows that the same DMC-EKF empirical-acceleration channel recovers a known matched acceleration vector to 2.15% relative error and lowers RMSE relative to EKF when all stations are visible and measurement noise is intentionally low. We report this as a *predeclared Cartesian-structural-channel bounded negative*: the standard structural response to dynamics bias does not transfer to this sparse-visibility regime under the predeclared rule, so the audit returns a bounded negative on both the noise-adaptive family (AUKF, PUKF) and the Cartesian structural-channel family (DMC-EKF) on the higher-fidelity slice. This strengthens the audit’s scope: the compact-model AUKF mechanism diagnostic is not rescued by the canonical Cartesian structural channel above the compact- J_2 ceiling under sparse visibility.

5.7 Predeclared parametric structural-channel response (DSA-EKF)

DSA-EKF gives a statistically detectable but below-floor improvement on the 40-minute higher-fidelity slice, so it remains a predeclared negative. One diagnostic concern for the Cartesian channel of Section 5.6 is that a generic per-axis empirical acceleration may be the less specific structural channel for the actual mismatch (drag/density/ballistic-coefficient bias); the standard LEO structural recipe estimates the multiplicative *drag-scale* parameter directly [77, 86, 46]; learned per-satellite drag-coefficient estimation has also been pursued for OD and prediction [8], whereas the diagnostic here shows a parametric drag-scale channel is inert under sparse visibility. S-Table 27 reports the predeclared seven-state DSA-EKF on the same higher-fidelity slice as DMC-EKF: the augmented state is $[r, v, \beta]$ with β a first-order Gauss–Markov drag-scale parameter about its nominal value of one, and the deterministic flow scales the drag acceleration by $\beta(t)$ on top of the otherwise compact two-body+ J_2 +drag model. The Gauss–Markov hyperparameters were selected on a disjoint validation seed over a four-point predeclared grid before the held-out test; the selection record and timestamped rule are retained for inspection. The predeclared positive criterion (DSA-EKF strictly lowest in mean, strictly negative paired CI versus the best non-DSA comparator, gap above the 3% practical significance floor defined in the statistical protocol) is *not* satisfied: DSA-EKF gives the lowest mean (347.67 m versus EKF 354.69 m), and the DSA-EKF–EKF paired-bootstrap CI is strictly below zero (-7.0 m , $[-9.7, -4.8] \text{ m}$), but the absolute gap lies *below the predeclared 10.6 m floor*. The associated Wilcoxon signed-rank descriptive p -value on the same pairing is highly significant ($p \approx 1.8 \times 10^{-9}$); we report this as an explicit illustration of the manuscript’s predeclared posture that *statistical detectability is not the same as practical significance* — a strictly negative paired CI versus the EKF coexists here with a reported, statistically robust within-floor improvement of about 7 m, and the predeclared rule therefore returns a bounded negative regardless of the underlying p -value. A below-floor strictly-negative CI of this kind does not imply that the structural channel provides *zero* benefit; below-floor is not the same as inert, but it is also not a positive contribution under the predeclared rule because the measured benefit is smaller than the practical significance threshold. The estimated drag-scale parameter moves by only $|\hat{\beta} - 1| \sim 10^{-7}$ from its prior over the 40-minute filter window, confirming that the parametric channel is essentially unobservable from sparse line-of-sight measurements at this window length: insufficient drag-induced orbit decay accumulates in 40 min to constrain β . We report

Table 6: Main structural-channel implementation sanity check. The favorable-geometry diagnostic is not a primary endpoint and not an operational OD case; it brackets the sparse-visibility negatives by showing that the implemented structural channels can recover matched known signals when the measurement geometry is intentionally easy.

Channel	Known signal	Last-window estimate	EKF→channel RMSE [m]	Readout
DSA-EKF drag scale	$\beta = 4.00$	$\hat{\beta} = 3.981$; 0.46% error	0.72→0.71	Matched beta channel recovers under all-visible, low-noise geometry.
DMC-EKF acceleration	empirical $[2.00, -1.00, 1.50] \times 10^{-5}$	$[2.02, -1.05, 1.50] \times 10^{-5}$; 2.15% error	4.20→3.45	Matched acceleration channel recovers under the same sanity regime.

this as a *predeclared parametric-structural-channel bounded negative* alongside the Cartesian DMC bounded negative, so the audit closes both standard structural responses to dynamics bias under sparse visibility, not only one.

5.8 Structural-channel sanity and drag-scale cascade

Favorable-geometry checks show the structural channels can recover matched signals, while the three-slice cascade still returns bounded negatives and localizes the drag-scale failure modes. Before interpreting the sparse-visibility structural-channel negatives, Table 6 records the favorable-geometry sanity check that the implemented DSA-EKF drag-scale channel and DMC-EKF empirical-acceleration channel can recover matched known signals under intentionally easy measurements. This is not a primary endpoint and not an operational OD result; it is a positive implementation check showing that the later negatives are not caused by a structural-channel implementation that cannot move under any geometry.

Table 7 then gives the three-slice drag-scale mechanism cascade. The first slice fixes the structural form to the true multiplicative drag-scale bias and identifies EKF analytic linearisation as the binding failure mode for the EKF-based candidate. The second slice substitutes deterministic sigma-point propagation of the augmented flow, removes that failure mode, and leaves a separate sparse-visibility observability-side bound: DSA-UKF is near UKF/AUKF in RMSE but $\hat{\beta}$ remains effectively at one. The third slice enriches the geometry and extends the arc; even with 9.6 km median arc-mean drag-induced separation, median trajectory-mean $\hat{\beta} = 1.0000$ and median maximum $|\hat{\beta} - 1| = 0.0010$, isolating candidate-side β prior inertia while also stressing the unaugmented UKF-family baselines. We therefore read the third row as a candidate-side beta-inertia diagnostic plus a long-arc sigma-point-stress record, not as a well-conditioned UKF-family ranking. This three-way mechanism separation—linearisation, observability, and candidate-side prior inertia—is the paper’s central non-self-referential diagnostic: it localises the structural-channel failure to a specific binding constraint rather than re-measuring a known mode. The inertia is consistent with an identifiability argument: the line-of-sight innovation is absorbed by the well-observed position/velocity correction while β enters only through twice-integrated drag, so the minimum-variance update attributes a position offset to r, v rather than to β , and the favorable-geometry sanity check (Table 6) recovers β precisely because that aliasing is suppressed. The full grid and stability caveats are in supplementary Section 8.

5.9 Long-arc higher-fidelity force-and-density-mismatch replication

Extending the arc to three hours does not rescue DSA-EKF; it is still worse than AUKF and the drag-scale state moves only marginally.

A practical concern on the 40-minute higher-fidelity slice is that the drag-scale structural channel may simply remain unobservable over such a short window. We therefore run a *long-arc higher-fidelity force-and-density-mismatch replication* (Table 8; S-Table 29) with a 3-hour arc, richer truth-side force and density mismatch, and the same compact estimator-side dynamics. DSA-EKF hyperparameters are selected on a disjoint long-arc validation seed before held-out testing.

The decision floor is an approximate astrodynamics-grounded Cramér–Rao position floor under the configured geometry and stated independence assumptions. The supplement gives the derivation and

Table 7: Main mechanism cascade for the drag-scale structural channel. Positive candidate-minus-reference gaps mean the candidate is worse than the best eligible reference. All three rows use held-out test populations after the relevant selection rule is frozen; none satisfies its predeclared positive criterion.

Slice	Observed-step comparison [m]	RMSE	Candidate—reference [m] (95% CI)	Drag-scale response	Mechanism readout
Matched channel, EKF predict	DSA-EKF UKF 359.7	1648.5	vs +1288.9 +2561.4]	[+378.1, max $ \hat{\beta}-1 = 0.0012$; final $\hat{\beta} = 1.0000$	Structural form is matched, but the EKF predict step is the binding failure mode.
Sigma-point substitution	DSA-UKF AUKF 313.1	316.4	vs +3.3 [−7.5, +16.2]	max $ \hat{\beta}-1 = 0.0009$; final $\hat{\beta} = 1.0000$	Sigma-point propagation removes the EKF divergence but not the AUKF guardrail.
Geometry-enriched DSA-UKF	DSA-UKF EKF 742.7	4879.9	vs +4137.2 +11235.7]	[−827.0, max $ \hat{\beta}-1 = 0.0010$; final $\hat{\beta} = 1.0000$	Drag signal is visible (9595.9 m mean, 29294.1 m final), but estimated $\hat{\beta}$ remains inert; UKF-family baselines are long-arc stressed, so this row is not a stable UKF ranking.

Table 8: Long-arc higher-fidelity force-and-density-mismatch result. DSA-EKF fails its predeclared positive criterion because AUKF is the best non-DSA estimator and the DSA-EKF—AUKF interval is positive. The auxiliary EKF/AUKF stress observation is retained only as supplementary stress context.

Quantity	EKF	UKF	AUKF	PUKF	DMC-EKF	DSA-EKF
Mean observed-step RMSE [m]	577.05	345.05	312.66	405.98	577.05	597.09

Held-out $n = 64$. DSA-EKF—AUKF mean +284.4 m, CI [+8.7, +697.9] m. The predeclared CRLB floor is 106.0 m and treats visible steps as independent; this can tighten the floor under within-pass correlation. The supplement reports a 36.8–335.1 m sensitivity range; pass-correlated S-Table row F is the opposite-direction bound at 335.1 m. Decision unchanged because DSA-EKF fails direction before floor magnitude under every variant.

sensitivity audit using Tapley–Schutz–Born statistical OD theory [77]. The resulting 106.0 m floor is unchanged when range-rate noise is added as a worst-case perturbation check, so the floor is not being set by an omitted range-rate term.

The substantive result is negative for transfer: even on the longer, richer slice DSA-EKF remains worse than AUKF, and the drag-scale state still shows only marginal movement. We therefore read this slice as a higher-fidelity long-arc failure-to-transfer result, not as evidence that the compact- J_2 EKF-over-AUKF ordering generalizes. The detailed interval and stability audit remains in the supplement.

5.10 KalmanNet linear-canonical sanity reproduction

The official KalmanNet linear-canonical sanity check (S-Table 36) reproduces the published release [64] at reduced scale and confirms the trained KalmanNet sits within 0.2 dB of the optimal Kalman filter on its own native benchmark. The load-bearing learned-claim comparator for this paper’s bounded negative remains the in-house KalmanNet-style learned-gain comparator (S-Table 78). The manuscript also reports a separately predeclared KalmanNet SPOT-OD transposition under four documented adaptations—orbital-scale normalization, full-arc curriculum rematching, sparse-observation masking, and lower-rate budget recalibration—as a feasibility note for readers who would otherwise ask whether the external architecture was tried at all (supplementary Section 9, S-Tables 8–9). Under that transposition, the upstream gain network remains far above the best classical reference on the held-out SPOT-OD test population even after a six-fold budget extension from 300 to 1800 optimizer steps: mean observed-step RMSE descends from 152.7 km to 136.0 km, while the best classical reference (AUKF) remains 340.9 m and the validation-best 1200-step snapshot still sits at 139.1 km. Because that gap is still consistent with an undertrained or mismatched transposition, we report it only as bounded feasibility evidence for this documented adaptation and budget schedule, not as load-bearing evidence for the paper’s learned-negative claim and not as an architecture-level conclusion.

Table 9: Powered DBAR characterization and withdrawal. DBAR does not beat the no-information majority classifier after 450 independently seeded realizations, so the heuristic is withdrawn as a claimed positive and retained only as a characterized negative.

Regime family	N	DBAR fires	R -adaptation bad	Accuracy
Nominal control	150	0	14	90.7%
Measurement-noise stress	150	10	6	89.3%
Dynamics/force-model bias	150	104	64	65.3%
All	450	114	84	81.8%

The always-no-fire majority baseline is 81.3%; DBAR’s +0.4-point increment has Wilson 95% interval [78.0, 85.1]% containing the baseline. The predeclared rule is therefore not a validated classifier.

5.11 DBAR (predeclared heuristic, withdrawn)

DBAR is withdrawn because the powered simulator characterization does not exceed its no-information baseline and the compact public SP3 replay does not transfer.

The mechanism diagnostic motivated a predeclared decision rule for the Dynamics-Bias Adaptation-Risk (DBAR) indicator. On the powered 450-realization simulator characterization, DBAR is not statistically distinguishable from its 81.3% no-information majority baseline because the Wilson 95% interval still contains that baseline.

The public precise-reference probe is also negative. The original ten-arc LAGEOS diagnostic reaches only 6/10 correct, below the 0.80 always-no-fire majority baseline. The stronger result is the formal 400-arc compact recursive-filter replay: 400 arcs were attempted, 373 completed, and 27 are carried as explicit status records rather than hidden or imputed (17 insufficient observations; 10 public product unavailable or non-parseable).

On the 373 completed arcs, DBAR is correct on 256/373 arcs (0.686), below the 0.737 no-fire majority baseline, with sensitivity 0.480 and specificity 0.760. Pooled held-out RMSE is EKF 566.60 m, fixed-noise UKF 573.24 m, AUKF 601.38 m, and SP3-IC propagation 779.45 m. Under the first-minus-second convention, EKF-minus-AUKF has mean -34.78 m, median -0.76 m, and 95% CI $[-71.84, -2.76]$ m, while fixed-noise-UKF-minus-AUKF has mean -28.14 m, median -0.77 m, and CI $[-61.80, -2.75]$ m; both intervals therefore indicate larger held-out RMSE for AUKF on average.

The practical interpretation is narrow. DBAR remains a useful predeclared stress test because it produced a clear negative and forced explicit withdrawal, but it is not promoted as a transferable decision statistic. Table 9 and Section 6 define the final scope.

5.12 Multi-revolution SGP4-reference benchmark

The public-catalog multi-revolution stress keeps recursive filters at kilometre scale and the learned residual worse than the best classical filter. The *multi-revolution SGP4-reference state-error benchmark with real ground-station geometry* (five real public LEO objects, SGP4-propagated reference from archived CelesTrak TLEs, 12-hour 7.07–7.81-revolution arcs, sparse line-of-sight measurements from the real eight-station geometry, every estimator on the compact two-body+ J_2 +drag OD model, so *the SGP4 reference is a genuine multi-revolution dynamics-model mismatch*; S-Table 43) shows the recursive classical filters at kilometre scale (fixed-noise UKF *best at 1809.54 m and best on three of five targets*, AUKF 1968.29 m, EKF 2052.35 m), the hard-bounded learned residual *worse at 2755.43 m and best on none of the five targets*, and the offline batch fit far worse (26256.53 m). This SGP4 slice is a bounded multi-revolution model-mismatch and public-catalog provenance stress only (Results-scope convention).

5.13 Public precise-reference provenance and measurement-pipeline integration

Public SP3/ILRS probes reduce provenance and measurement-pipeline risk, include one predeclared prospective public-week readout with a numerically lower learned residual mean but confidence interval spanning zero,

and still do not validate the simulator result.

This public precise-reference layer is a bounded provenance and measurement-pipeline stress layer, not estimator-skill evidence. Recent ILRS-supervised TLE/state-vector refinement and hybrid SGP4 error-learning studies motivate explicit split, baseline, and scope guardrails for any learned public-reference claim [3, 69]. Accordingly, the present paper treats the public layer as a transfer-risk diagnostic rather than as proof that ML cannot help on real SLR.

The direct public readouts now separate into three bounded roles. First, on real LAGEOS-1/-2 ILRS normal points against independent ILRS NSGF SP3-c products, the original ten-arc state-error slice places the compact recursive filters at hundreds of metres (fixed-noise UKF 334.72 m, AUKF 341.33 m, EKF 367.98 m), and the DBAR public check reaches only 6/10 correct. Second, the formal 400-arc replay summarized in Section 5.11 remains the strongest compact transfer diagnostic for the adaptive-filter warning: 373 completed arcs clear the project’s completed-arc sizing target, yet DBAR stays below its no-information baseline and AUKF is worse than EKF and UKF on mean paired differences. Third, the predeclared prospective public-week readout with the full correction stack records an indeterminate numerical ordering: validation selected UKF (full correction) at 426.53 m, but on the once-scored public week 260523 the learned residual UKF reaches 365.45 m versus 385.29 m for the best recursive classical candidate, a learned-minus-best-recursive-classical gap of -19.84 m with confidence interval spanning zero and learned lower on 4/10 completed test arcs (S-Table 56).

The broader pipeline probes show breadth without upgrading the claim. The SP3/CRD breadth probe covers 10 targets, 40/40 target-weeks, 240 fixed SP3 start epochs, 25,449 CRD normal points, and 29 stations; all-target and non-LAGEOS intervals cross zero, while the held-out test stratum remains positive in favour of the higher-fidelity model. The range-residual case study on four LAGEOS arcs and 384 normal points gives pooled held-out RMS 93.09 m for classical range-only WLS, 207.49 m for the learned residual on the SGP4 prior, and 96.08 m for WLS plus learned calibration, with learned estimators best on 0/4 arcs.

These public measurements therefore reduce data-handling and provenance risk for future validation and record one indeterminate predeclared public-week readout under the full correction stack, but they still do not validate the compact simulator result or establish stable real-data learned superiority. Operational ILRS precise-orbit analyses reach centimetre to sub-metre accuracy with full reductions; the roughly 350–925 m compact-model regime here remains about three orders of magnitude away from that ceiling.

5.14 Practical significance floor and slice discrimination

The floor and slice-discrimination audits preserve the learned negative and keep the nominal split out of the discriminative claim set. The 15-seed cohort observed-step gain on the all-step propagation-dominated reference (S-Table 58) agrees in direction with the primary observed-step endpoint above: the *15-seed aggregate stress gain* of RGR-GF over UKF is 355.78 m, with *all 15 seed aggregates favoring RGR-GF*, leading with direction (15/15) and magnitude not a p -value, and the same diagnostic is negative versus AUKF (two of 15 seeds). These magnitudes share direction but not estimand; the reconciliation is consolidated in S-Table 82 (Appendix 15).

Endpoint-support draw. The $K = 8$ endpoint-support draw is a seed-disjoint check under the selected observed-step endpoint, not external or pre-training-cohort preregistration; Section 4.5 gives the timeline and evidentiary boundary. On that fixed- n draw the per-scenario best method is the tuned AUKF on the nominal and stress splits and the causal EKF on the controlled mismatch; no learned estimator is the per-scenario best on any split, so the rule returns *no learned positive on any scenario at the primary endpoint*. The paired RGR-GF-minus-best-classical observed-step gaps at $K = 8$ are nominal +9.4 m, CI [+1.1, +21.7] m; stress +44.8 m, CI [−0.2, +87.1] m; controlled mismatch +48.7 m, CI [+19.1, +97.1] m. The central evidence anchor is the later frozen-rule replication reported in Table 4.

Discriminative and non-discriminative slices. The stress and controlled-mismatch outcomes carry the discriminative weight because their observed effects are above the practical decision floor and remain directionally negative for the learned estimator in the $K = 32$ frozen-rule replication under the established endpoint hierarchy (Table 4). The stress row is additionally checked by the stress-only $K = 96$ replication, which exceeds the $K \approx 94$ floor-power design check for approximately 0.80 power at the 27.5 m stress-floor

effect and still returns no learned positive. The nominal split is retained for traceability but is *formally de-prioritized* from the set of discriminative endpoints: its learned-versus-classical spread is at or below the practical floor and is power-unresolved in the endpoint-fixation record, so it is practically non-discriminative regardless of the later post-hoc sensitivity. Discriminative conclusions are drawn only from the stress and controlled-mismatch slices, the controlled and higher-fidelity force-mismatch slices, the multi-revolution SGP4-reference benchmark, the tail-conditioned dense-tracking probe, and the predeclared structural and noise-adaptive channels (Sections 5.5, 5.6, 5.7).

Floor sensitivity. The *practical significance floor sensitivity sweep* (1, 2, 3, 5%) is in S-Table 59: at every swept floor the headline finding is preserved, so the predeclared 3% choice is *not* load-bearing for the negative. This 3% practical-significance floor is a study-specific audit threshold, not a domain-conventional standard; a mission-specific OD campaign would replace it with its own acceptance threshold. The value was fixed in the supplementary evidence materials before the independent endpoint-fixation draw was run; its timestamped record is included for inspection.

5.15 Less-constrained learned residual and residual-scale sweep

Relaxing the residual bound does not create a learned positive; the less-constrained and larger-scale residual checks fail across the tested scenarios. A possible concern is that the central negative could be *preordained by the bound*: the canonical residual is hard-bounded (0.03 scaled-unit tanh) and additionally damped by a learned gate, context budget, and prior-anchoring penalty, so any competitiveness with the classical prior bank may be inherited “by construction”. We test this with a *less-constrained learned residual comparator* (RGR-U, S-Table 60): identical architecture/capacity/inputs/curriculum, but the residual budget, gate, context budget, and prior-anchoring/activity/entropy/visibility training penalties *all removed*. Removing the bound does *not* produce a learned advantage: the paired RGR-U-minus-best-classical observed-step gap exceeds seven kilometres on *every* scenario. This divergence is consistent with sparse-visibility training instability for unbounded residuals and should not be read as a clean single-factor ablation of only the residual bound. To address whether the canonical-vs-unconstrained gap is binary or graded, we additionally trained two *intermediate residual-scale* comparators (tethered $s = 0.30$ and tethered $s = 1.00$); the residual-scale sweep (S-Table 61) still shows graded collapse in this fixed architecture and budget, with the gap growing monotonically with the residual scale on the tethered branch and becoming an order of magnitude larger on the untethered endpoint. Taken together, RGR-U and the residual-scale sweep establish only a graded, construction- and budget-conditioned collapse rather than a purely binary failure; they do not establish a clean single-factor bound ablation or architecture-independent monotonicity.

This compact-simulator negative is informative but not externally decisive. It would be self-fulfilling if the audit only tested a tightly bounded residual against its own prior in sparse geometry. Instead, the record also shows that roughly four-fifths of evaluated steps are zero-visible, that an offline WLS reference can use the visible arc much more effectively, that RGR-U and residual-scale checks fail when residual bounds and anchoring are relaxed, and that favorable all-visible structural-channel controls can recover matched signals. The drag-scale cascade then adds a separate structural readout: even with a 9.6 km median arc-mean drag-induced separation, the geometry-enriched DSA-UKF slice leaves the drag-scale estimate near its prior, while the DSA-EKF/DSA-UKF cascade separates EKF linearisation, observability, and candidate-inertia limits. These findings do not establish that learned OD corrections fail under richer physics or different sensors, but they do show that the tested sparse-geometry/prior-inertia construction fails after less-bounded residual and structural-channel escape routes were audited.

6 Claim Audit

This section records only the interpretation boundaries, since the numerical results are stated in Section 5 and the full row-level tables are in the supplement. Table 2 gives the compact claim–evidence–limit hierarchy; the full endpoint timeline is defined once in Section 4.5.

Within that boundary, the claim is narrow: no evaluated learned construction selected without test-set information beats the per-scenario best tuned classical reference on the primary observed-step endpoint, and

the endpoint-choice sensitivity audit does not change that conclusion; PUKF, DMC-EKF, and DSA-EKF fail their own predeclared positive criteria; and DBAR is withdrawn because the powered characterization does not exceed its no-information baseline or transfer to the public SP3 check. The conclusion would change only if a learned construction passed the primary endpoint, capacity/input-matched controls, practical floor, and multiplicity audit in the documented setting, or if a structural channel satisfied its predeclared positive rule on the higher-fidelity or long-arc population. External validity would require a new measurement setting with its own force/noise model, references, baselines, and predeclared gates rather than reinterpretation of the current simulator tables.

The $K = 32$ frozen-rule replication has limited sensitivity to an exactly floor-sized stress effect (approximate power ≈ 0.42 at the 27.5 m floor under a one-sided $\alpha = 0.05$ t -approximation; under the predicate’s stricter two-sided 95% CI rule the realized floor power is somewhat lower), so power is reported at the predeclared floor effect rather than at the realized gap; with the realized stress-row paired-gap variance, about $K \approx 94$ independent realizations would be needed for roughly 0.80 power at that floor. The stress-only $K = 96$ internal replication exceeds that design check for approximately 0.80 power at the 27.5 m floor and preserves the negative, but it remains internal, simulator-bound evidence under the already selected endpoint.

7 Limitations and Missing Experiments

Results are evaluation evidence for the documented simulation protocol, windows, baselines, and reported tables. They do not validate the estimator for flight operations, real-time integrated testbed deployment, safety-critical autonomy, or all sparse-visibility regimes. Only a single canonical untuned stochastic EnKF boundary check is reported (S-Table 4); it does not displace the classical baselines on any slice, but tuned, inflated, or localized EnKF variants and sequential Monte Carlo OD methods remain outside this comparator set, so the bounded ensemble-Kalman negative should not be read as addressing the broader ensemble-filter design space. Likewise, cislunar particle/Gaussian-mixture filters and coupled orbit-attitude/sensor-tasking frameworks are cited as adjacent operational/filtering context, not as comparator baselines addressed by the tested claims. Graph message passing remains a controlled diagnostic rather than an isolated causal contribution—the visible-station mean-deviation aggregator reduces to the identity whenever a single station is visible, so cross-station message passing is a structural no-op at exactly the zero- and single-visible-station steps that dominate the evaluation window, which is why the capacity-matched controls cross zero rather than indicating that relational structure is unhelpful in principle—and the auxiliary replay, controlled-synthetic, and bounded public precise-reference slices are stress/provenance probes rather than operational OD.

The self-audit demonstration is a self-evaluation: the record is designed by the authors and applied primarily to the authors’ own residual constructions, plus in-house learned-gain and adapted-transposition probes. That structure can falsify the authors’ own claims before escalation, but it is not independent third-party validation of a reusable framework and does not show that another group would instantiate the same seven ingredients or reach the same boundaries. The novelty-audit definitions are self-authored operational definitions: inspectable and falsifiable per cell, but not an externally ratified evaluation taxonomy. This negative is architecture-family specific to the tested constructions and compact sparse-visibility setting. The present contribution is a worked, inspectable compact-simulator audit record; independent application by a separate research group is the next evidentiary tier before any generally validated reusable-framework claim.

The post-hoc endpoint-timeline concern qualifies the primary endpoint ranking evidence; it does not drive the AUKF R -inflation or drag-scale cascade mechanism findings, and the DBAR withdrawal follows its own characterization rather than learned endpoint selection. The designated next step for this specific limitation is an externally timestamped digest of a prospective decision rule, committed before and bound to an independent public precise-reference replication carried out by a separate party, with no rule content disclosed at timestamping; this is future work, neither completed here nor claimed as external trial-registry preregistration.

The public LAGEOS/SP3 compact-filter readout is only a bounded real-measurement adaptive-filter transfer diagnostic for the DBAR/AUKF R -inflation warning: the statistic is computed from the real AUKF-versus-fixed-UKF innovation and R -adaptation stream, while the outcome label is defined externally from held-out SP3 state error, not from the DBAR statistic and not from a simulator label (S-Tables 45 and 46).

The original ten-arc slice, extended 80-arc replay, and formal 210-arc replay remain archival evidence; the formal 400-arc replay supersedes the earlier campaigns as the strongest compact real-data diagnostic. It attempted 400 arcs; 373 completed; the 27 non-completed arcs are explicit status records in the validation records (17 insufficient observations; 10 public product unavailable or non-parseable), not hidden or imputed. The completed count clears the earlier approximate 185 comparable-arc formal power target (comparable arc: completed arc with at least 10 CRD normal points and SP3 state coverage), but the result remains negative. DBAR is correct on 256/373 arcs (0.686), below the 0.737 always-“no-fire” majority baseline, with sensitivity 0.480 and specificity 0.760; EKF-minus-AUKF mean -34.78 m has median -0.76 m and 95% CI $[-71.84, -2.76]$ m, and fixed-noise-UKF-minus-AUKF mean -28.14 m has median -0.77 m and CI $[-61.80, -2.75]$ m. Under the first-minus-second convention, those strictly negative intervals mean AUKF has larger held-out RMSE on average than EKF and fixed-noise UKF. The supplementary completed-only stratification shows non-stable ordering across preceding/train/validation/test splits and across LAGEOS-1/LAGEOS-2; the wider formal 400-arc intervals are therefore best treated as epoch, target, arc-geometry, and unmodelled perturbation/frame-reduction heterogeneity rather than stable filter superiority, with no causal attribution. The larger replay therefore supports withdrawing DBAR and treating the compact-model R -inflation mechanism as a hypothesis for future real-data studies; it does not validate the simulator conclusion, establish operational POD, or provide real-data learned estimator-skill evidence.

The simulator omits several failure modes that bound interpretation of the negative results. It does not include maneuvers, thrust events, attitude-coupled force errors, clock-estimation states, sensor timing biases, heterogeneous station instrumentation, decoded RF observables, onboard GNSS ambiguity handling, or a multi-class operational tracking network. Its estimator-side dynamics deliberately stay below an operational force-model ceiling; the long-arc and public probes test selected pressure points, not the full space of high-fidelity force, sensor, and reference-model errors. Consequently, a negative in this paper weighs against advancing the tested claims within the documented sparse-visibility setting, but it does not transfer as a negative about a different sensor geometry, mission class, or operational POD system.

The compact-physics ceiling is also a mechanism limitation. A learned above-floor positive requires residuals that are jointly *observable* from the visible measurements and *representable* by the candidate relative to the prior/filter bank. Each evaluated negative fails at least one gate: on the noise-side stress split the signal is already represented by AUKF, while on the force-side mismatch and drag-scale cascade it is unobservable from sparse line-of-sight geometry and/or unrepresentable by the compact prior. This is descriptive of the tested constructions, not a general impossibility result. Offline WLS, multi-revolution public-catalog, and public precise-reference probes show that compact dynamics and sparse geometry dominate large parts of the residual budget, so the result is a pre-campaign evidence-triage negative for the tested constructions under this compact prior, not a claim that learned correction could never help under a different force model, sensor, or public-reference regime.

Before operational deployment, future validation must bridge from compact-sim gates (learned candidates surpass tuned adaptive guardrails, frozen endpoints, practical floors, structural channels) to higher-fidelity simulation, real held-out arcs with independent precise reference, and operations-facing metrics. This study covers only compact-sim gates; escalation remains resource-triage future work, not operational POD. The long-arc EKF/AUKF reversal shows compact-AUKF prescription is regime-bound.

Missing experiments are concrete: centimetre-level precise SLR reduction; real high-fidelity force-error campaigns; maneuvers; powered, independently sized reference-state scoring of learned estimators on real measurements beyond the bounded public probes; decoded RF measurements; independent external confirmation or operational validation of the endpoint result; and faithful reproduction of specific published learned-OD systems in their own documented settings. Highest-priority follow-ups are topology generalization to held-out satellites, stations, and geometries; high-fidelity force/sensor mismatch beyond the current controlled split; selector generalization on held-out public campaigns; broader non-graph parameter-matched ablations with public precise-reference validation; and independent public precise-reference validation beyond replay-style evidence.

Public replay scope and deployment boundary. The robustness claims are limited to the synthetic measurement, visibility, and noise regimes summarized in the reported tables and figures. Under the time-aligned SatNOGS observation-window replay (Appendix 15) the EKF stays finite at kilometre scale and offline robust batch WLS produces an accurate postfit OD solution, so that slice is a bounded classical/offline-OD

validation, not a universal-failure slice or confirmatory deployment evidence [41].

8 Data and Reproducibility Statement

The evaluation suite is designed to be reproducible from archived inputs, fixed random seeds, documented simulator settings, and reported result tables. Archived Celestrak and SatNOGS snapshots replace live API queries, so replay scenarios are not affected by later public-service changes. The public precise-reference probes use real ILRS satellite-laser-ranging normal points for LAGEOS-1 and LAGEOS-2 together with independent ILRS NSGF SP3-c products; consistent with Section 6, they are provenance and measurement-pipeline evidence under the claim hierarchy, not operational POD or real-data learned-estimator validation. Synthetic nominal and stress splits use fixed orbit, force-model, visibility, and measurement-noise settings, and all methods are evaluated over a common scoring horizon.

The versioned supplementary evidence package supplied with the submission records the fixed data settings, input and output digests, predeclared rules, seeds, splits, selected models, validation criteria, and evaluation records used by the reported tables. The evidence file and manifest are part of the confidential inspection materials supplied through the journal submission system, while the supplement quotes the primary endpoint, $K = 32$ frozen-rule, and stress-only $K = 96$ rule records so that the core predeclaration language is visible in the manuscript package rather than only described. Completed inspection checks now include four passing validation surfaces already archived in the repository: `results/validation/submit_validation_validation.md` (compiled-package readback, citation pass, release-packet pass, zero LaTeX-warning count), `results/validation/archive_extracted_reproduction.md` (archived-set integrity plus one public precise-reference OD slice rerun), `results/validation/minimum_tier_reproduction_check.md` (manifest, digest, archive-membership, and claim-map coverage audit), and `results/validation/targeted_retraining_replay_public.md` (one bounded from-scratch learned-estimator replay on deterministic materialized slices). The current active-regeneration harness separately verifies 9 generated table inputs plus 1 figure include from `paper/main.tex`, with 10/10 active artifacts passing and no mismatches or blockers. In the archive-extracted rerun, the recorded public CRD/SP3 inputs recompute the range-only filters and reproduce the submitted table text exactly while matching the recorded public claim summary with zero field mismatches. These checks materially strengthen inspection and rerun readiness, but they remain below full scientific reproduction. The evidence-package access boundary is stated in Data availability.

A prepared GitLab worktree and retained git metadata are provenance aids, not by themselves a public reproducibility guarantee for the current evidence package. The package becomes publicly reproducible only when the evidence bundle, generated artifacts, manifests, and digests are committed to an accessible repository or deposited in a persistent archive. Until that public/persistent archive exists, or until a separate prospective external validation is completed, the manuscript remains capped at an inspectable internal-evidence tier rather than a fully public or independently validated reproduction tier.

One bounded learned-estimator training replay is included on deterministic slices of existing retained data. The replay runs the full materialized curriculum shape on 160 nominal training, 32 nominal validation, 96 stress-training, 24 stress-validation, and 16 SatNOGS replay-validation trajectories; it records finite stage histories, checkpoint and manifest digests, and an unchanged canonical checkpoint digest while leaving the submitted trained-estimator records unchanged. In parallel, the official KalmanNet release is rerun end-to-end on its native linear-canonical benchmark inside the evidence package and recovers the expected ordering observation-floor $>$ KalmanNet $\geq$ optimal KF, with KalmanNet test error within 2 dB of the optimal KF on that upstream benchmark. These are bounded inspection artifacts, not full paper-table reproduction, a full seed-suite rerun, or a main-result reproduction. A full scientific rerun would require regenerating the benchmark data, retraining the learned estimators under the stated curriculum and seeds, evaluating all classical and learned methods on the nominal, stress, and replay scenarios, and rebuilding the displayed figures and tables.

9 Conclusion

The primary finding is a compact-simulator AUKF R-inflation diagnostic: under force-model mismatch in a compact two-body+ J_2 +drag OD simulator, AUKF inflates effective R (mean 3.32; median R -only

NIS 4.73 versus 1.60/1.83 for EKF/UKF) and damps needed state corrections, quantifying an OD-specific instance of the known covariance-matching failure mode. The drag-scale cascade separates EKF linearisation, sparse-geometry observability, and candidate-inertia limits as distinct binding constraints. The mechanism result is reproduced in the controlled diagnostic and is further supported by the R-inflation cap control (S-Table 28); the supplement’s long-arc EKF/AUKF stress observation confines the compact- J_2 prescription to its tested regime. These compact-simulator findings (350–925 m scale) require precise-reference validation before operational transfer; operational ILRS/SP3 analyses operate at centimetre/sub-metre accuracy under higher-fidelity reductions, a scale not addressed here.

The bounded negative on learned estimators is secondary and exploratory: no evaluated learned construction beats the per-scenario best tuned classical reference on the primary observed-step endpoint, including the stress-only $K = 96$ floor-power-scale check for approximately 0.80 power at the 27.5 m stress-floor effect. The endpoint-choice sensitivity audit finds no learned positive under either observed-step or all-step scoring, and the structural-channel and DBAR diagnostics fail their predeclared positive criteria. Because the endpoint was selected after training-cohort review and the $K = 32/K = 96$ records are internal frozen-rule replications rather than external preregistration (Section 4.5), the finding remains a bounded screening result that still needs prospective external confirmation. It is architecture-family specific to the bounded tanh-residual, prior-anchored RGR-GF family and tested comparators. The separately predeclared upstream KalmanNet transposition is retained only as feasibility context for one documented external architecture and budget schedule; because it remains 136–153 km against a 340.9 m AUKF reference after the bounded six-fold extension, it is not treated as load-bearing evidence for the learned-negative claim. RGR-U and residual-scale checks limit, but do not eliminate, the concern that bound removal or prior anchoring contributes to the in-house result.

The transferable element is the audit discipline, not the empirical result. Other OD regimes would need their own endpoint hierarchy, frozen decision rules, classical and offline guardrails, capacity-matched controls, practical floors, evidence records, datasets, baselines, and external-reference checks. The present contribution is a worked, inspectable compact-simulator audit and campaign-triage record; independent external application is the next evidentiary tier before any generally validated reusable-framework claim.

Campaign-planning workflow. For small-satellite teams screening learned candidates, the evidence record supports a compact-simulator gate workflow: candidates must beat the per-scenario best tuned adaptive guardrail on observed-step RMSE by at least the practical floor, survive paired-bootstrap CIs and frozen decision rules under $K=32$ -scale replications, move a structural drag-scale channel under sparse visibility, avoid provenance false-positive triggers, and then escalate to precise-reference validation. In the current record, RGR-GF and the structural channels fail those gates: RGR-GF loses to AUKF on stress (gap +31.9 m after powered $K=96$) and to EKF on force-mismatch (gap +16.8 m after $K=32$), DSA-EKF remains below practical floor, and DBAR withdraws after 450-realization characterization. The practical implication is therefore narrow but clear: within this compact screening regime, AUKF remains the adaptive guardrail to beat before any higher-fidelity campaign is justified. Confirmatory use still requires prospectively fixed external validation.

10 Reproduction checklist

Every headline is inspectable from the supplementary evidence materials supplied with the submission. The completed checks are deliberately separated and already logged in repository validation artifacts: `submission_validation.md` verifies compiled-PDF readback, citations, release-packet synchronization, and zero-warning LaTeX logs; `active_manuscript_regeneration.md` verifies 9 generated table inputs plus 1 figure include from the current `paper/main.tex` with 10/10 active artifacts passing and no mismatches or blockers; `archive_extracted_reproduction.md` verifies archived-set integrity and reruns one public precise-reference OD slice from recorded CRD/SP3 inputs; `minimum_tier_reproduction_check.md` validates manifest inventory, digests, review-archive membership, and claim-map coverage without rerunning estimators; `targeted_retraining_replay_public.md` records one bounded learned-estimator replay with finite objective histories, checkpoint digests, and unchanged canonical submitted checkpoints. The upstream KalmanNet release is also rerun on its native benchmark as an architecture-sanity comparator, and the bounded EnKF

comparator is rerun to verify that its negative remains bounded after regeneration. A deposit-ready minimal public artifact bundle has also been prepared: manuscript sources, generated tables and figures, synthetic-split generation scripts, rule records, seeds, archived public-input provenance, validation reports, and per-artifact SHA-256 digests are already assembled so that a persistent public archive can be minted without repackaging. These checks support review inspection under the established scope hierarchy. They remain below full scientific reproduction: full learned-estimator training, full seed-suite regeneration, all-filter recomputation, and live public-data retrieval are deeper rerun routes that can require hours to days, and the public slice remains a measurement-pipeline/provenance check rather than operational POD validation. The supplement uses standalone S-table/S-figure labels and an independent reference list for the separately submitted supplementary file; if production requests a merged reference list, only citation formatting changes are required, not scientific content. Data availability defines the current confidential access boundary and the committed public-release path.

Each headline therefore points to recorded input-data digests, predeclared rule records, and records that fix seeds, splits, trained models, and selection rules. The exact confidential access mechanism is stated once in Data availability; this checklist maps claims to the records needed for inspection or deeper reruns.

Supplementary material

The complete learned-estimator architecture and training-objective specification, the bounding and diagnostic tables that support but are not central to the claim-audited negative, and the four temporal/calibration figures are reported in the separate supplementary document that accompanies this manuscript. The supplement provides full printed table bodies for core and boundary material (the multiplicity-adjusted family, the DBAR three-regime exhibition and 450-realization characterization, the densified-visibility control, and the matched-budget and SatNOGS-selection sensitivity tables); selected lower-priority archival records (precise-reference measurement-pipeline tables plus propagation/offline-OD/replay reference tables) appear as concise indexed summaries in the supplement, with their full numeric records supplied in the confidential journal-review evidence package. Cross-references from the main text to supplement-housed tables and figures use the S-Table/S-Figure prefix and resolve through the supplement’s external label list.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The confidential journal-submission package includes versioned dependency and evaluation records sufficient for inspection of the reported retained results. The confidential inspection materials supplied through the journal submission system include the versioned supplementary evidence package and manifest for the current submission version. They cover fixed data settings, internally timestamped predeclared rule and seed records, trained-estimator records, evaluation records, learned-estimator specification records, recorded public input identifiers, and retained digest values used in the study. The main manuscript and supplement expose the claim hierarchy, table bodies or indexed summaries, rule excerpts, and key numeric results; the confidential package is needed to inspect retained digests, full selected lower-priority numeric records, trained-estimator and evaluation records, and inspection-material extraction checks. In parallel, a deposit-ready minimal public artifact bundle has been prepared from the current version: manuscript sources, generated tables and figures, synthetic-split generation scripts, table-regeneration and validation harnesses, pinned rule

records, seeds, archived public-input provenance, and per-artifact SHA-256 digests are already assembled for persistent archival deposition at the first venue-permitted public release point. The submitted evidence demonstrates internal record checks, generated-table consistency from retained records before and after inspection-material extraction, one public precise-reference OD slice-level rerun from submitted inspection materials, one bounded learned-estimator training replay, one official KalmanNet native-benchmark rerun, and one regenerated bounded EnKF comparator. These completed checks strengthen inspection readiness but do not yet constitute full scientific reproduction: full raw-data generation, full estimator retraining, all recursive-filter recomputation, and live public-data retrieval remain deeper rerun routes beyond the present submission package. No DOI or public identifier is claimed at initial submission because that identifier can only be cited after the archive mints it; once assigned, it will be inserted into the manuscript metadata and release manifest without altering the scientific record.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used AI-assisted technologies to support prose organization, clarity revision, and consistency verification across the manuscript and supplementary materials. After using these tools, the authors reviewed and edited all content and take full responsibility for the scientific claims, evidence interpretation, and published content.

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